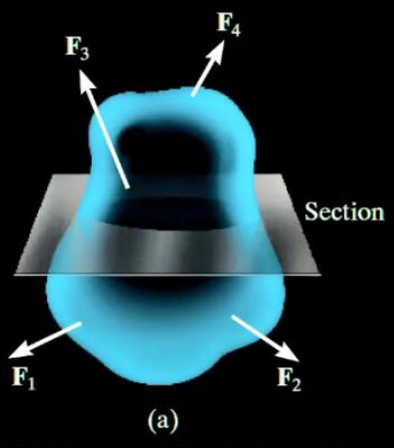
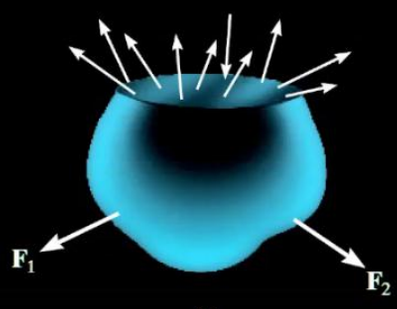
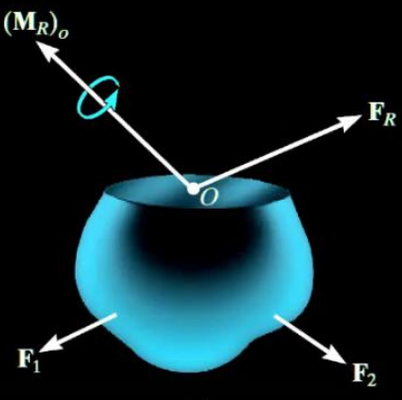
Internal Resultant Loadings On Solid Cross Sections:



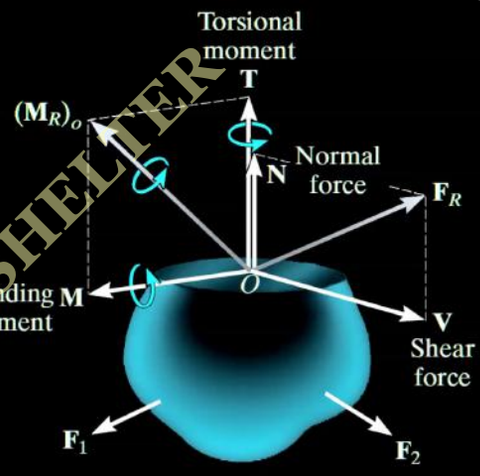
(a)



(b)



(c)



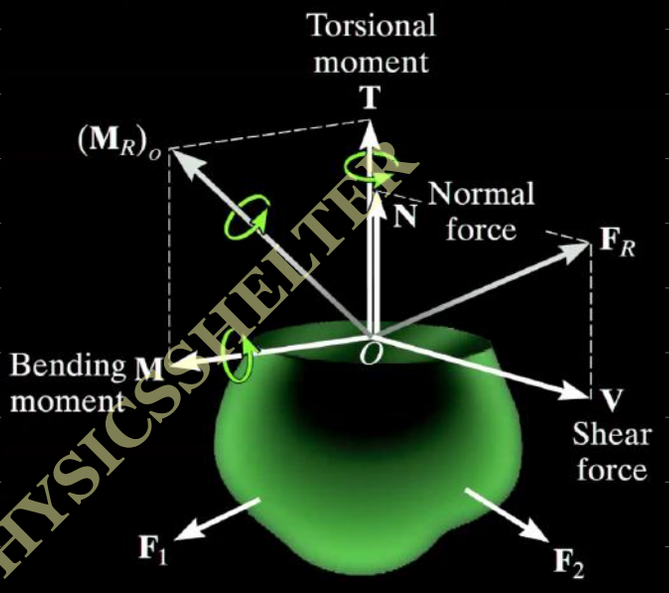
(d)

When any engineer designs a product, very often he is interested in knowing the internal profile of forces. Just analysing various components as rigid bodies may not be enough because whether a material will fail or where a material is likely to fail depends on the profile of internal forces and torques acting within the matrix of the material. In this chapter we will be mostly dealing with such internal force and torque profiles across various cross-sections of any engineering component. For instance consider the body in figure (a) above which is in equilibrium under action of four external forces. To obtain internal loadings acting on specific regions within the body, we pass an imaginary section or "cut" through the region where internal loadings are to be determined. The two parts of the body are then separated and a free body diagram of one of the parts is drawn. Here there is a distribution of internal forces acting on the exposed area or section. These forces represent the effects of the top section of the body on the bottom section. Although exact distribution of this internal loading may be unknown, we can find its resultant force and torque (F_R & M_R) at any specific point O by applying equations of equilibrium

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to the bottom part of the body.
The point O is often chosen as centroid of the area.
In case of long and slender bodies, the section to be considered is generally taken perpendicular to the longitudinal axis of the body. This section is referred to as cross section.

General Load Components On A Section:



Various components of forces and torques can be obtained by considering the equilibrium of sectional FBD. Various load components acting on the section are:

Normal Force (N): This force acts perpendicular to the area. It is developed when external forces tend to push or pull the two segments of the body.

Shear Force (V): The shear force lies in the plane of the area and it is developed when the external loads tend to cause the two segments of the body to slip over one another.

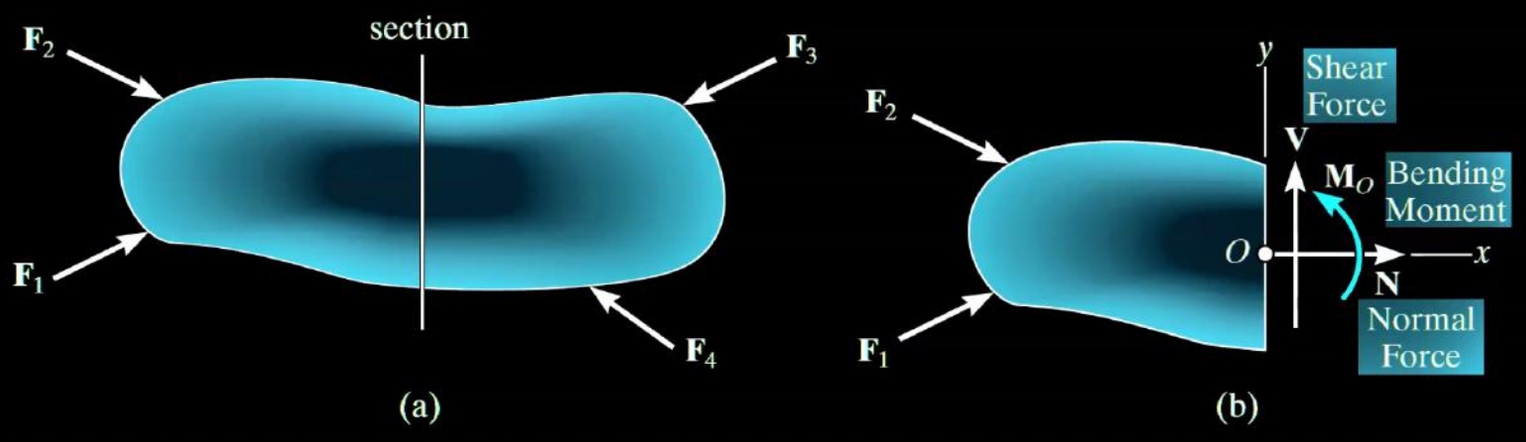
Torsional Moment (Torque) T: This effect is developed when external loads tend to twist one segment of the body with respect to the other about an axis perpendicular to the area

Bending Moment: (M): The bending moments are caused when external loads tend to bend the body about an axis lying within plane of the area.

→ All the four types of loads on the section are shown in the figure above for a clear visualization.

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COPLANAR LOADINGS:



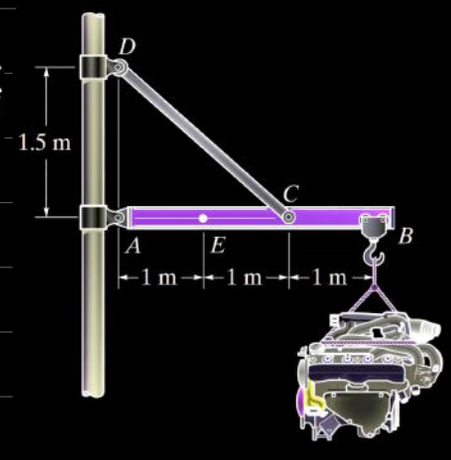
A loading is said to be coplanar if all the points on which the force is applied lie in the same plane and the forces are also acting along that plane. In case of coplanar loading, twisting moment is not present and we only have shear force, Bending moment and normal reaction. If we use x, y and z coordinate axes as shown on the left segment, we can obtain N by applying $\Sigma F_x = 0$, V by applying $\Sigma F_y = 0$ and Bending moment M_o can be obtained by balancing torques about point O (z axis) which avoids moments caused by unknowns N and V.

→ Before finding out internal loadings at any section, it might be necessary to first find out external loadings on any solid member. This can be done using standard methods of rigid body statics.

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EXAMPLE:

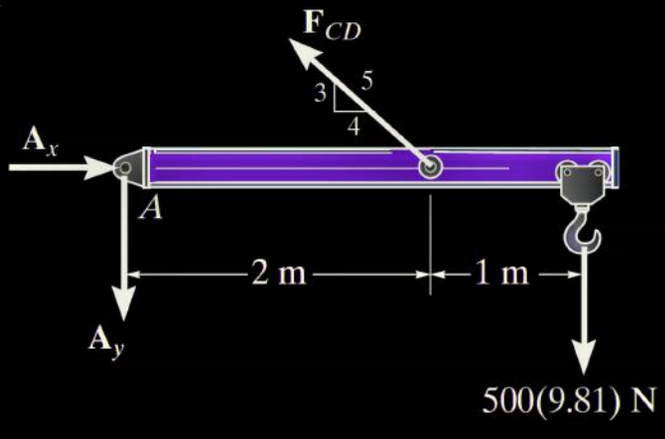
The 500-kg engine is suspended from the crane boom in Figure. Determine the resultant internal loadings acting on the cross section of the boom at point *E*.



Assume that except for the engine all other parts of the system are massless.

Solution:

Support reactions:



$$\circlearrowleft +\Sigma M_A = 0; \quad F_{CD}\left(\tfrac{3}{5}\right)(2\text{ m}) - [500(9.81)\text{ N}](3\text{ m}) = 0$$

$$F_{CD} = 12\,262.5\text{ N}$$

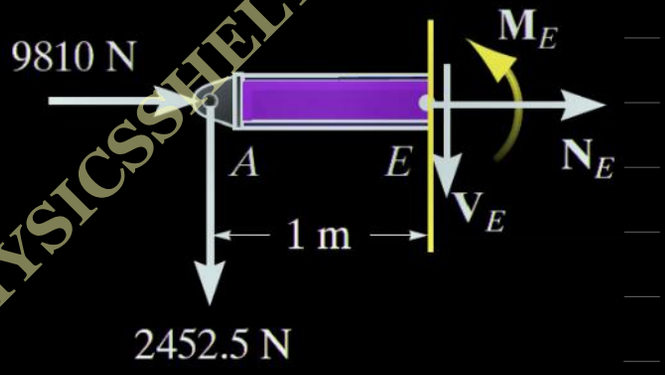
$$\pm\rightarrow \Sigma F_x = 0; \quad A_x - (12\,262.5\text{N})\left(\tfrac{4}{5}\right) = 0$$

$$A_x = 9810\text{ N}$$

$$+\uparrow \Sigma F_y = 0; \quad -A_y + (12\,262.5\text{ N})\left(\tfrac{3}{5}\right) - 500(9.81)\text{ N} = 0$$

$$A_y = 2452.5\text{ N}$$

Sectional FBD:



$$\pm\rightarrow \Sigma F_x = 0; \quad N_E + 9810\text{ N} = 0$$

$$N_E = -9810\text{ N} = -9.81\text{ kN} \quad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \quad -V_E - 2452.5\text{ N} = 0$$

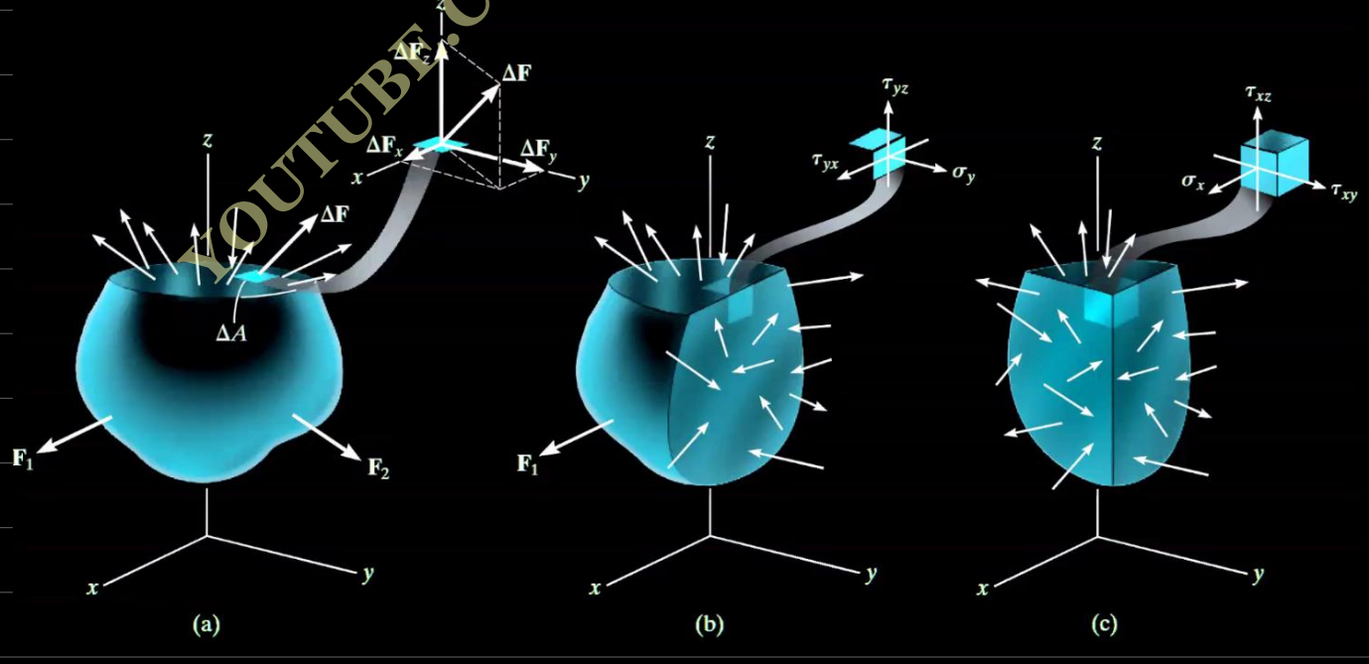
$$V_E = -2452.5\text{ N} = -2.45\text{ kN} \quad \textit{Ans.}$$

$$\circlearrowleft +\Sigma M_E = 0; \quad M_E + (2452.5\text{N})(1\text{ m}) = 0$$

$$M_E = -2452.5\text{ N}\cdot\text{m} = -2.45\text{ kN}\cdot\text{m} \quad \textit{Ans.}$$

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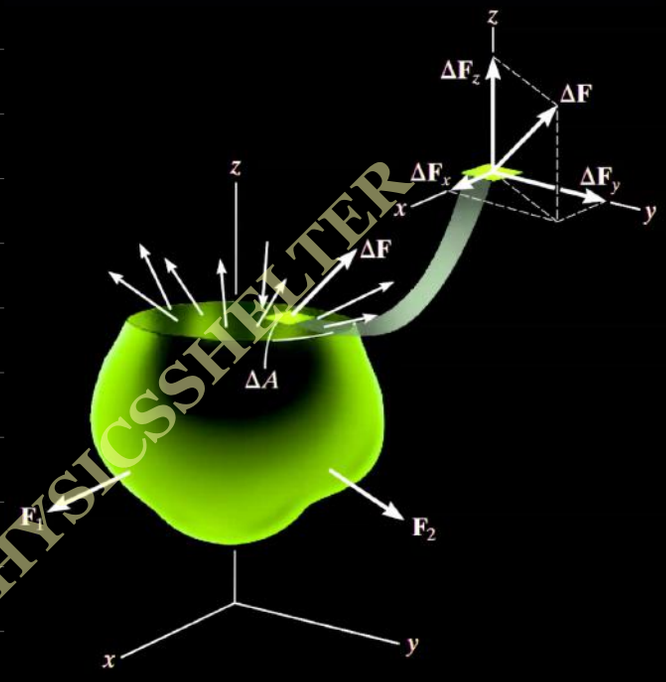
STRESS:



In the previous section we talked about resultant loading on

a section of a solid body. However we understand that actually the resultant loading is a simplification of the actual loading which is distributed over the entire cross section and for analysis of internal mechanics of materials we often need to focus on finding this distribution. To find this distribution, we will define the concept of stress.

Let us imagine the cut section area to be further divided in to small areas such as ΔA shown in adjacent figure. As we imagine ΔA to be smaller and smaller we make two assumptions i.e (a) Material is continuous (b) Material is cohesive (no cracks etc). As ΔA becomes smaller and smaller so does $\vec{\Delta F}$. We can resolve $\vec{\Delta F}$ in to three components : ΔF_x, ΔF_y and ΔF_z. Even though in the limit, the ΔA tends to zero and so also the components ΔF_x, ΔF_y and ΔF_z but the quotient of force components to the area approaches a finite value. The set of these values defines stress (local stress) on that patch. The stress defines intensity of the internal force acting on a specific plane (area) passing through a point.



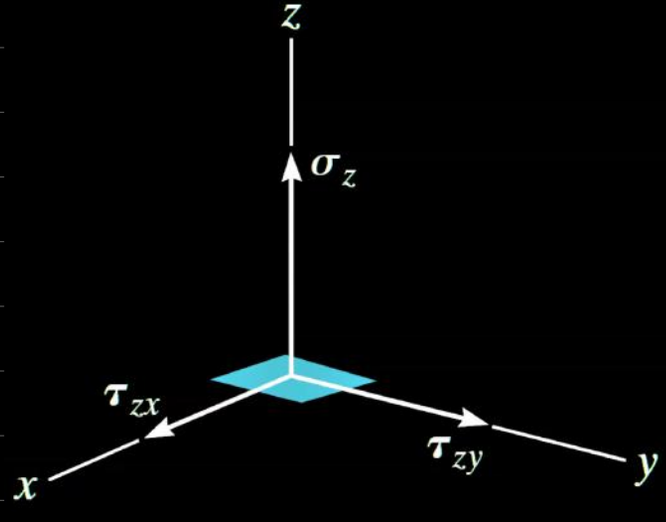
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NORMAL STRESS:

The intensity of force acting normal to ΔA is called normal stress.

Mathematically : $\sigma_z = \lim_{\Delta A \to 0} \frac{\Delta F_z}{\Delta A}$.

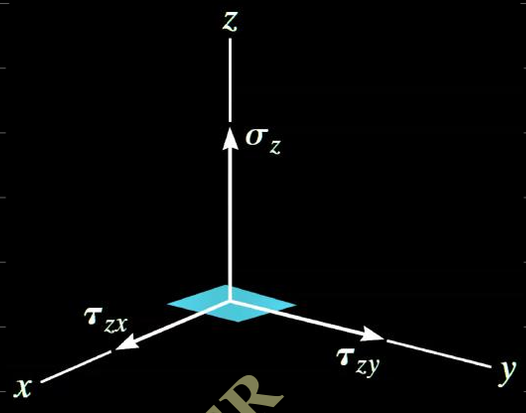
If normal stress pulls the patch, it is called tensile stress, if it pushes, it is called compressive stress.



SHEAR STRESS: The intensity of forces acting tangentially to the patch is called shear stress. Here we will have two components of shear stress:

$$\tau_{zx} = \lim_{\Delta A \to 0} \left(\frac{\Delta F_x}{\Delta A} \right)$$

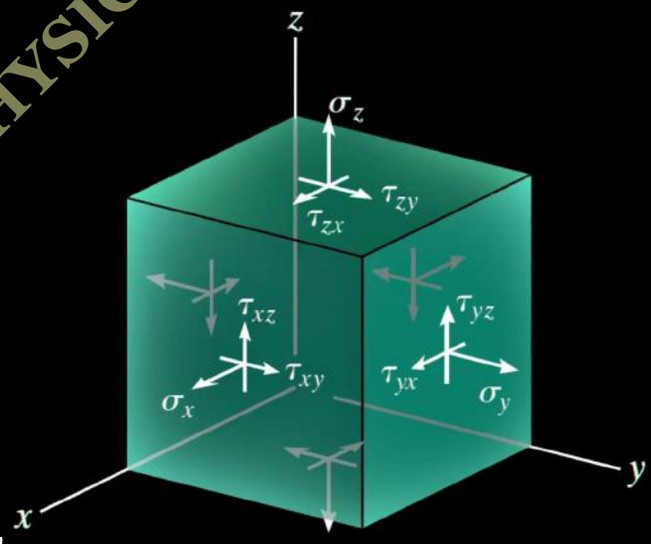
and $$\tau_{zy} = \lim_{\Delta A \to 0} \left(\frac{\Delta F_y}{\Delta A} \right)$$



The subscript z denotes the direction of normal to the area and x and y indicate the direction along which each shear stress acts.

GENERAL STATE OF STRESS:

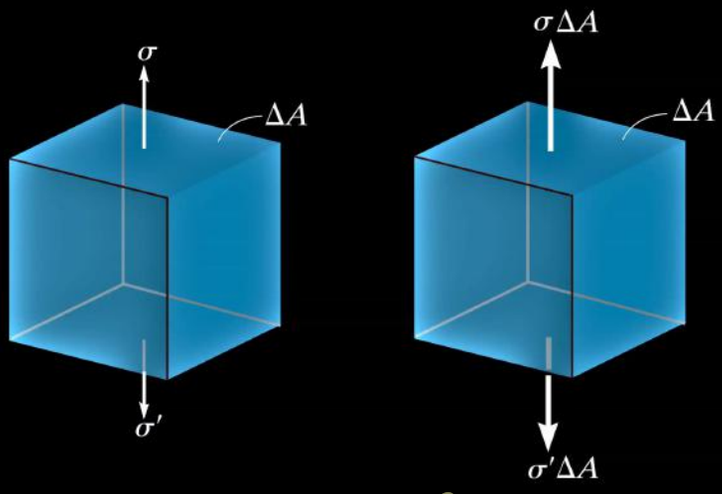
Just as we showed stresses on a patch with area normal along z axis, similarly we can think of patches with area normals along x and y axes. These patches at any point can be visualized to form the surface of an infinitesimal cuboid at a given point in a material. Each direction is characterized by one normal and two shear stresses. Thus in general stress at a point in solid is characterized by 9 values. When these values are written in a matrix, we call it stress tensor. (Just as vectors are represented by three components, Tensors in mechanics are represented by 3×3 matrices mostly. Though there can be tensors of higher orders also). Stress is measured in Pascals.



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Normal Stress Equilibrium:

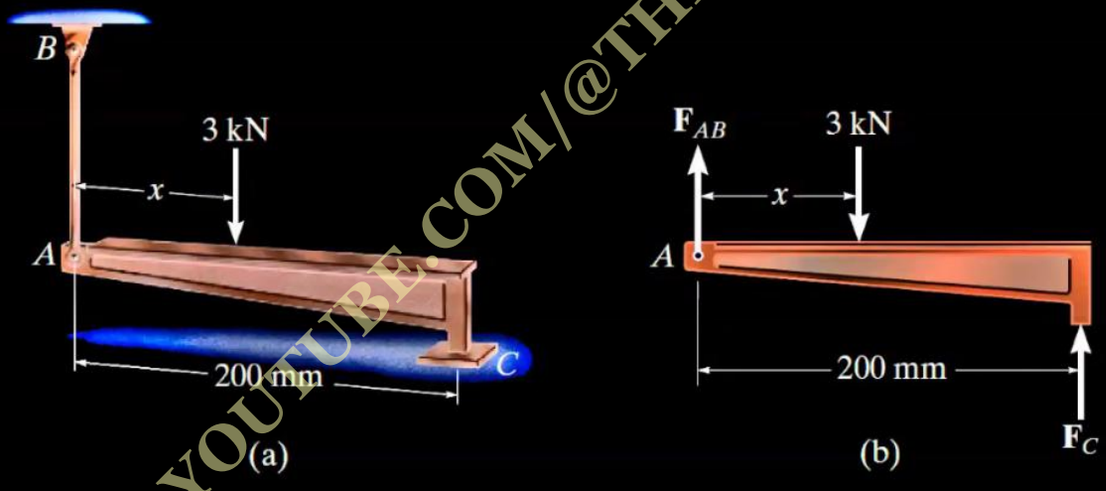
Considering FBD of a small cube shaped element we can readily see that for infinitesimal elements, the normal stress must be same on opposite faces. Thus in adjacent figure,
$\sigma \Delta A = \sigma' \Delta A \Rightarrow \sigma = \sigma'$.



Stress on element Free-body diagram

Example:

Member AC shown in Figure is subjected to a vertical force of 3 kN. Determine the position x of this force so that the average compressive stress at the smooth support C is equal to the average tensile stress in the tie rod AB. The rod has a cross-sectional area of 400 mm^2 and the contact area at C is 650 mm^2.



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SOLUTION

Internal Loading. The forces at A and C can be related by considering the free-body diagram of member AC, Figure b. There are three unknowns, namely, F_{AB}, F_C, and x. To solve we will work in units of newtons and millimeters.

$$+\uparrow \Sigma F_y = 0; \qquad F_{AB} + F_C - 3000\ \text{N} = 0 \qquad (1)$$

$$\circlearrowleft + \Sigma M_A = 0; \qquad -3000\ \text{N}(x) + F_C(200\ \text{mm}) = 0 \qquad (2)$$

Average Normal Stress. A necessary third equation can be written that requires the tensile stress in the bar AB and the compressive stress at C to be equivalent, i.e.,

$$\sigma = \frac{F_{AB}}{400\ \text{mm}^2} = \frac{F_C}{650\ \text{mm}^2}$$

$$F_C = 1.625F_{AB}$$

Substituting this into Eq. 1, solving for F_{AB}, then solving for F_C, we obtain

$$F_{AB} = 1143\ \text{N}$$

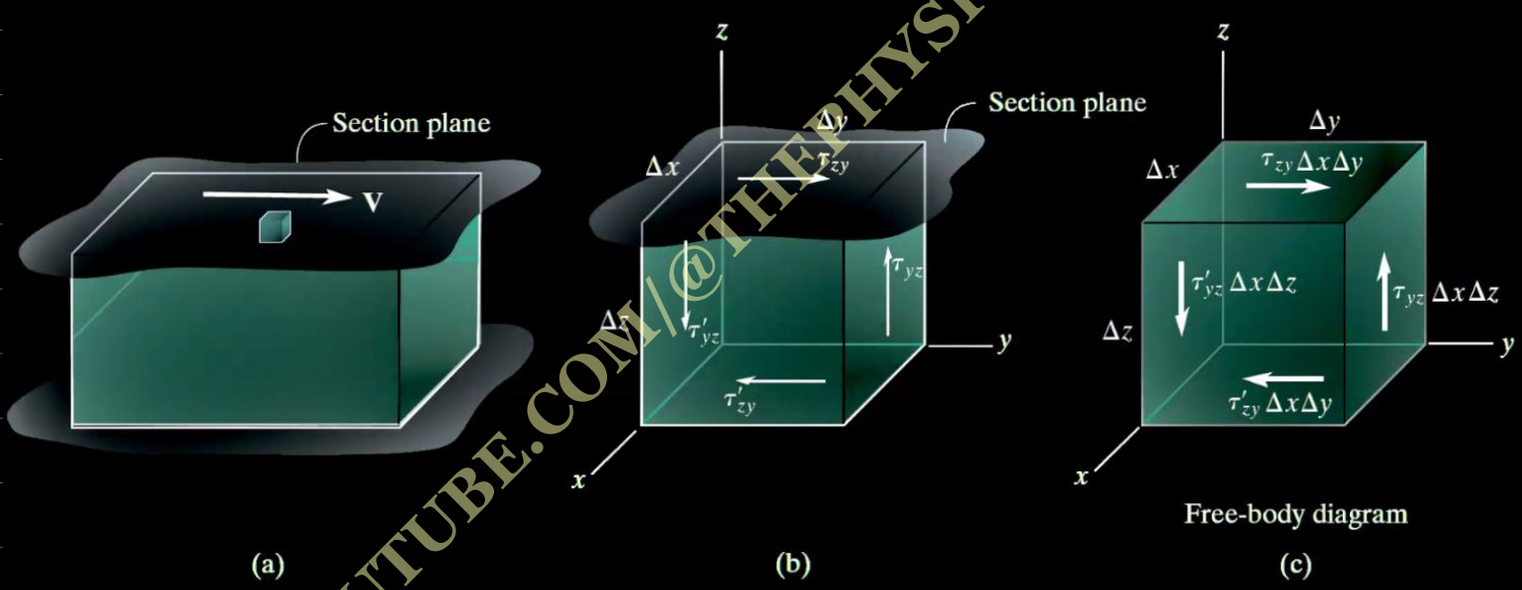
$$F_C = 1857\ \text{N}$$

The position of the applied load is determined from Eq. 2,

$$x = 124\ \text{mm} \qquad \textit{Ans.}$$

As required, $0 < x < 200$ mm.

Shear Stress Equilibrium:

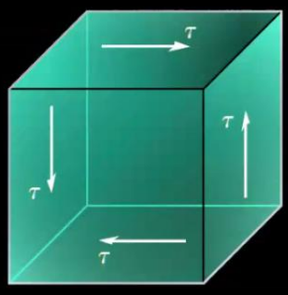


Shear Stress Equilibrium. Let us consider the block in Figure *a*, which has been sectioned and is subjected to the internal shear force V. A volume element taken at a point located on its surface will be subjected to a direct shear stress τ_{xy}, as shown in Figure *b*. However, force and moment equilibrium of this element will also require shear stress to be developed on three other sides of the element. To show this, it is first necessary to draw the free-body diagram of the element, Figure *c*. Then force equilibrium in the y direction requires

$$\Sigma F_y = 0; \qquad \underbrace{\tau_{zy}\underbrace{(\Delta x\ \Delta y)}_{\text{stress area}}}_{\text{force}} - \tau'_{zy}\,\Delta x\,\Delta y = 0$$

$$\tau_{zy} = \tau'_{zy}$$

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Pure shear

(d)

In a similar manner, force equilibrium in the z direction yields $\tau_{yz} = \tau'_{yz}$. Finally, taking moments about the x axis,

moment
force arm
stress area

$$\Sigma M_x = 0; \qquad -\tau_{zy}(\Delta x\,\Delta y)\,\Delta z + \tau_{yz}(\Delta x\,\Delta z)\,\Delta y = 0$$

$$\tau_{zy} = \tau_{yz}$$

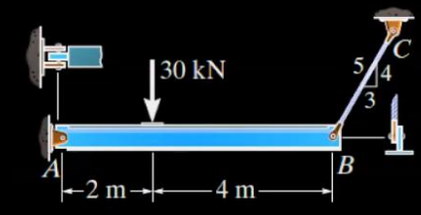
In other words,

$$\tau_{zy} = \tau'_{zy} = \tau_{yz} = \tau'_{yz} = \tau$$

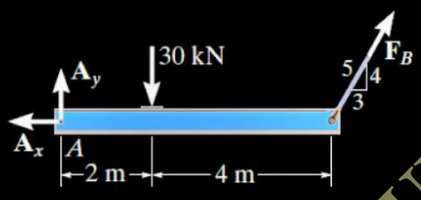
and so, ***all four shear stresses must have equal magnitude and be directed either toward or away from each other at opposite edges of the element***, Figure *d*. This is referred to as the ***complementary property of shear***, and the element in this case is subjected to *pure shear*.

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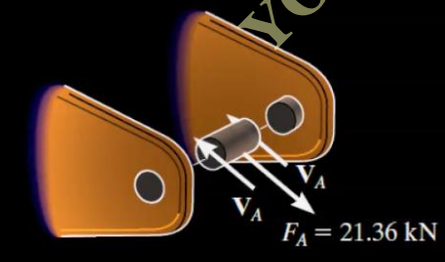
EXAMPLE



(a)



(b)



(c)

Determine the average shear stress in the 20-mm-diameter pin at A and the 30-mm-diameter pin at B that support the beam in Figure *a*.

SOLUTION

Internal Loadings. The forces on the pins can be obtained by considering the equilibrium of the beam, Figure *b*.

$$\circlearrowleft + \Sigma M_A = 0; \qquad F_B\left(\frac{4}{5}\right)(6\text{ m}) - 30\text{ kN}(2\text{ m}) = 0 \qquad F_B = 12.5\text{ kN}$$

$$\stackrel{+}{\rightarrow} \Sigma F_x = 0; \quad (12.5\text{ kN})\left(\frac{3}{5}\right) - A_x = 0 \qquad A_x = 7.50\text{ kN}$$

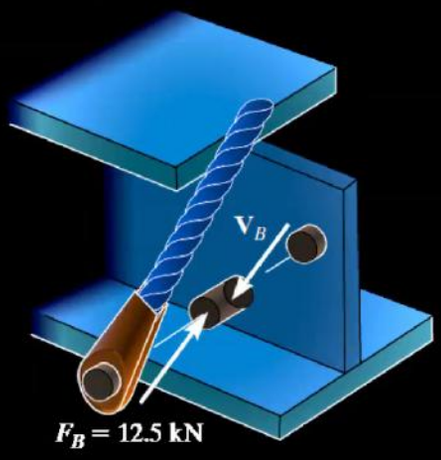
$$+\uparrow \Sigma F_y = 0; \quad A_y + (12.5\text{ kN})\left(\frac{4}{5}\right) - 30\text{ kN} = 0 \qquad A_y = 20\text{ kN}$$

Thus, the resultant force acting on pin A is

$$F_A = \sqrt{A_x^2 + A_y^2} = \sqrt{(7.50\text{ kN})^2 + (20\text{ kN})^2} = 21.36\text{ kN}$$

The pin at A is supported by two fixed "leaves" and so the free-body diagram of the center segment of the pin shown in Figure *c* has *two* shearing surfaces between the beam and each leaf. Since the force of the beam (21.36 kN) acting on the pin is supported by shear force on each of two surfaces, it is called ***double shear***. Thus,

$$V_A = \frac{F_A}{2} = \frac{21.36\text{ kN}}{2} = 10.68\text{ kN}$$



(d)

In Figure *a*, note that pin *B* is subjected to ***single shear***, which occurs on the section between the cable and beam, (Figure *d*). For this pin segment,

$$V_B = F_B = 12.5\ \text{kN}$$

Average Shear Stress.

$$(\tau_A)_{avg} = \frac{V_A}{A_A} = \frac{10.68(10^3)\ \text{N}}{\frac{\pi}{4}(0.02\ \text{m})^2} = 34.0\ \text{MPa} \qquad \textit{Ans.}$$

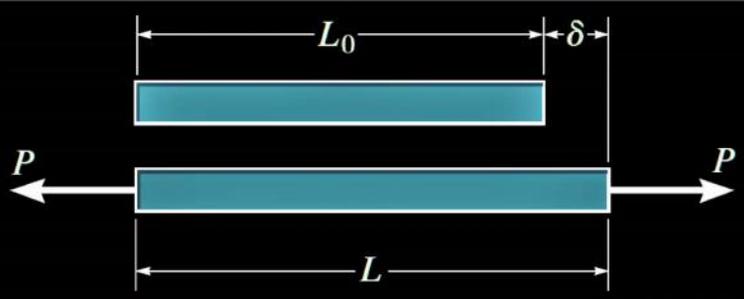
$$(\tau_B)_{avg} = \frac{V_B}{A_B} = \frac{12.5(10^3)\ \text{N}}{\frac{\pi}{4}(0.03\ \text{m})^2} = 17.7\ \text{MPa} \qquad \textit{Ans.}$$

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STRAIN: When various stresses act on the elements of a solid, the elements deform and the deformation can be described in terms of change in lengths of line segments within the body in various directions and also the changes in angles between various line segments. A quantitative description of such changes is defined using concept of strain.

LONGITUDINAL STRAIN:

If an axial load P is applied to the bar as shown in adjacent figure it will change Bar's length L_0 to L. We define the average normal strain (ϵ) of the bar as ratio of change in its length to the original length. Thus:

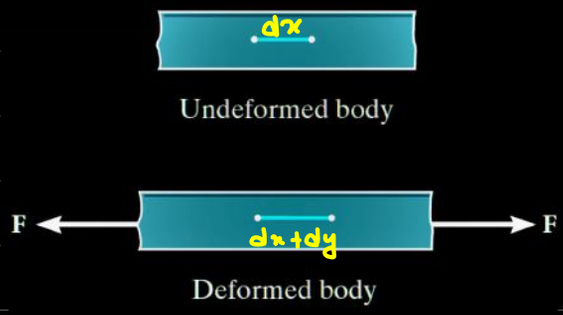


$$\epsilon_{avg} = \delta / L_0$$

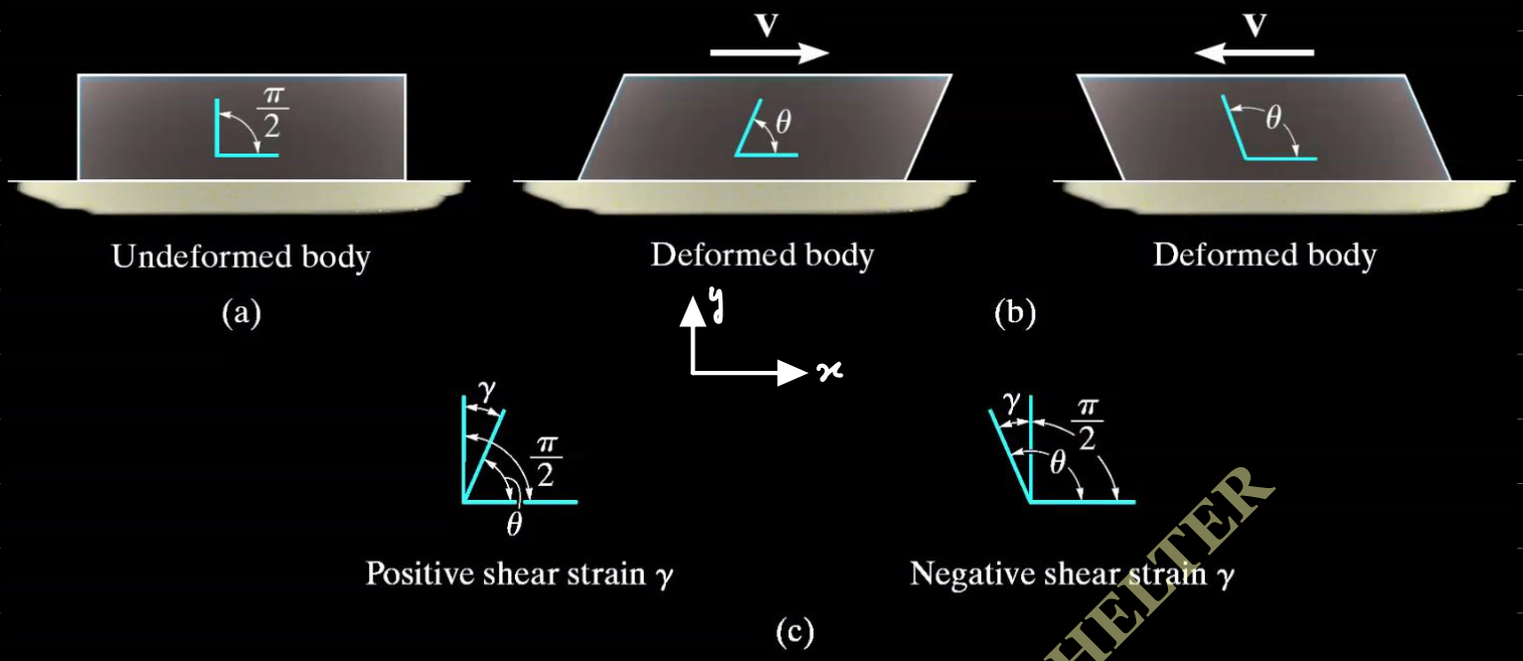
LOCAL LONGITUDINAL STRAIN:

If an infinitesimal element of length dx elongates by an amount dy then the local longitudinal strain is given as

$$\epsilon = dy/dx$$

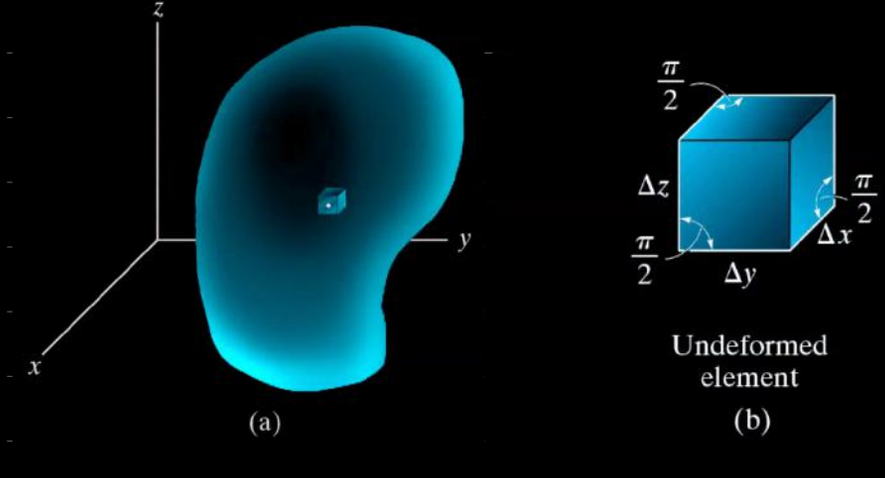


Shear Strain:



Deformations under action of various stresses not only cause elongations or contractions of line segments, they can also change the directions of various line segments. If we select two line segments that are originally perpendicular to each other (say along +x and +y directions) then the change in angle that occurs between them is called shear strain. For instance in above figures under action of applied load the angle between the two line segments becomes θ then the shear strain is $\boxed{\gamma = \pi/2 - \theta}$. Notice that if $\theta < \pi/2$ then shear strain is positive and if $\theta > \pi/2$, shear strain is negative.

Cartesian Strain Components:



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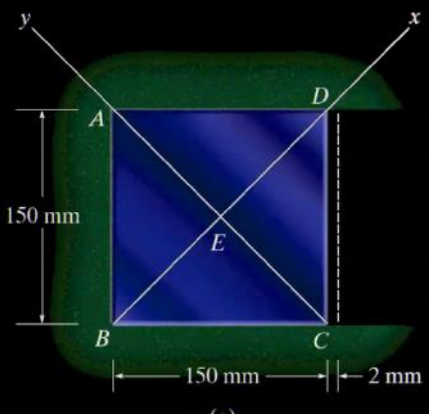
The definitions of normal strain and shear strain can be generalized to three dimensions. Consider an undeformed element at a point in a solid body (figure a). Since the dimensions of the element are very small, upon deformations, its shape will become a parallelopipe. Here normal strains change the sides of the element to $(1+\epsilon_x)\Delta x$, $(1+\epsilon_y)\Delta y$ and $(1+\epsilon_z)\Delta z$. Thus there may be change in volume of the solid element. The shear strains change the angles between the sides of the element to $\frac{\pi}{2}-\gamma_{xy}$, $\frac{\pi}{2}-\gamma_{yz}$ and $\frac{\pi}{2}-\gamma_{xz}$ thus producing a change in shape of the element.

SMALL STRAIN APPROXIMATIONS: In typical engineering applications, the strains occuring in the solids are very small i.e $\epsilon << 1$ and $\gamma << 1$. Under such conditions we may appropriately neglect second order terms containing products of strains and also we may assume $\gamma \approx \tan\gamma$ for purpose of easy calculations. (Note that γ must be always considered in radians). Also $\cos\gamma \approx 1$.

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EXAMPLE:

The plate shown in Figure *a* is fixed connected along *AB* and held in the horizontal guides at its top and bottom, *AD* and *BC*. If its right side *CD* is given a uniform horizontal displacement of 2 mm, determine (a) the average normal strain along the diagonal *AC*, and (b) the shear strain at *E* relative to the *x*, *y* axes.



(a)

SOLUTION

Part (a). When the plate is deformed, the diagonal *AC* becomes *AC'*, Figure *b*. The lengths of diagonals *AC* and *AC'* can be found from the Pythagorean theorem. We have

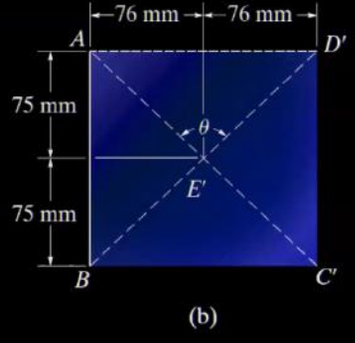
$$AC = \sqrt{(0.150\,\text{m})^2 + (0.150\,\text{m})^2} = 0.21213\,\text{m}$$

$$AC' = \sqrt{(0.150\,\text{m})^2 + (0.152\,\text{m})^2} = 0.21355\,\text{m}$$

Therefore the average normal strain along *AC* is

$$(\epsilon_{AC})_{\text{avg}} = \frac{AC' - AC}{AC} = \frac{0.21355\,\text{m} - 0.21213\,\text{m}}{0.21213\,\text{m}}$$

$$= 0.00669\,\text{mm/mm} \quad \textit{Ans.}$$



(b)

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Part (b). To find the shear strain at E relative to the x and y axes, which are 90° apart, it is necessary to find the change in the angle at E. The angle θ after deformation, Figure b, is

$$\tan\left(\frac{\theta}{2}\right) = \frac{76\text{ mm}}{75\text{ mm}}$$

$$\theta = 90.759° = \left(\frac{\pi}{180°}\right)(90.759°) = 1.58404\text{ rad}$$

Applying , the shear strain at E is therefore the change in the angle AED,

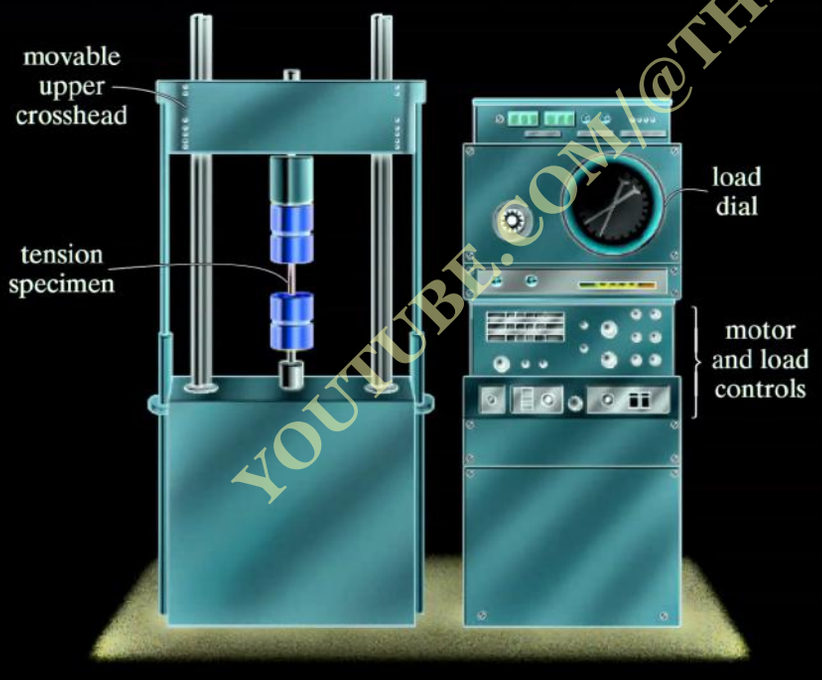
$$\gamma_{xy} = \frac{\pi}{2} - 1.58404\text{ rad} = -0.0132\text{ rad} \qquad \textit{Ans.}$$

The *negative sign* indicates that the once 90° angle becomes larger.

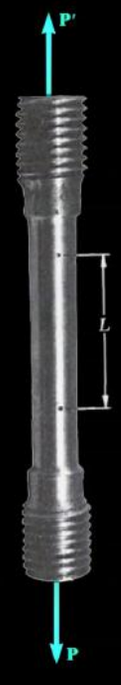
NOTE: If the x and y axes were horizontal and vertical at point E, then the 90° angle between these axes would not change due to the deformation, and so $\gamma_{xy} = 0$ at point E.

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STRESS STRAIN PLOTS UNDER LONGITUDINAL LOADING:



UNIVERSAL TESTING MACHINE



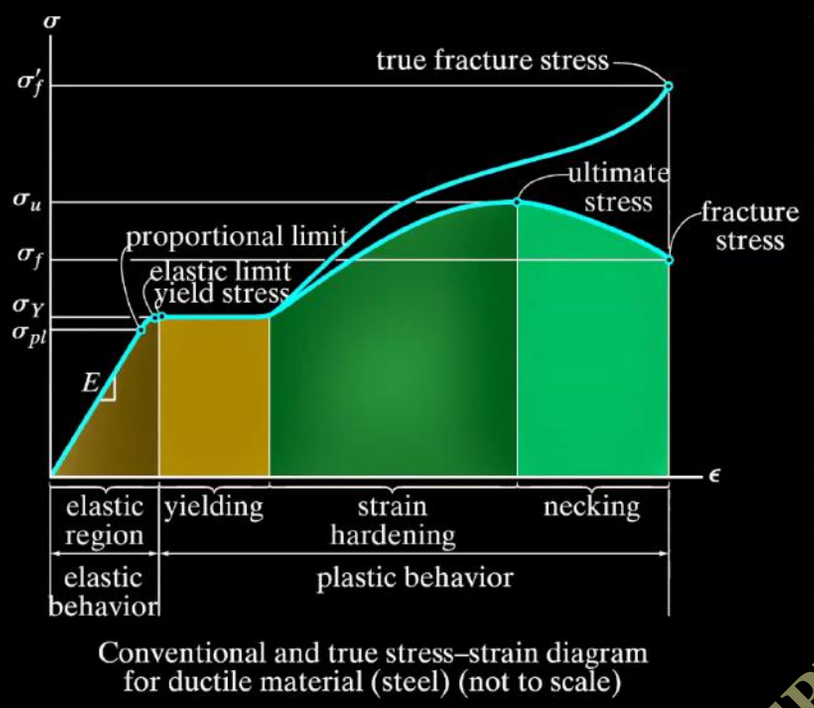
SPECIMEN (IRON BAR).

For testing of stress strain behaviour of solids, a universal testing machine is used whose sketch is shown above. A typical specimen to be tested is shaped such as the one shown in the right figure above. The specimen is gripped in the vices

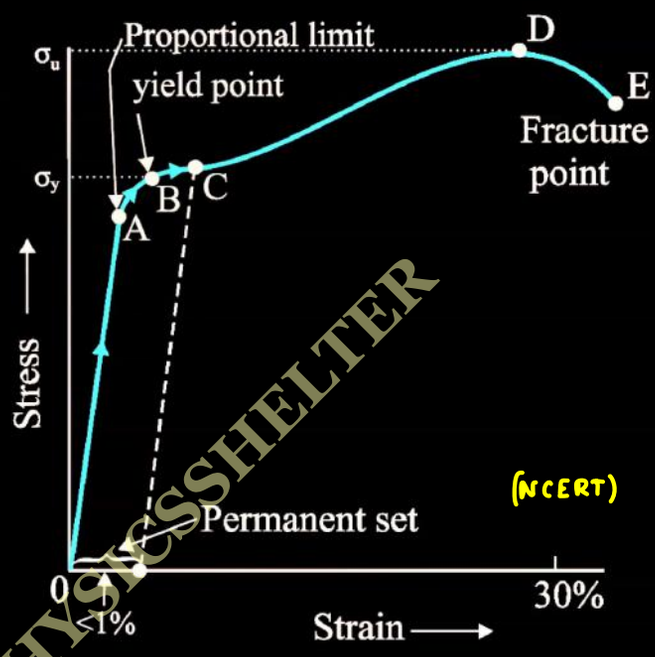
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at top and bottom and gradually elongated due to the vices moving apart. The experiment is typically continued until the bar fractures.

Loading Curves (Stress Vs Strain) For Various Materials:



Conventional and true stress–strain diagram for ductile material (steel) (not to scale)



A typical stress-strain curve for a metal.

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Engineering Stress (Nominal Stress) Vs True Stress: As a metal bar is stretched, its cross sectional area reduces. When the tension in the bar is divided by this reduced cross section we get true stress while if we divide the tension by original area, we get nominal stress or engineering stress. True stress plots are very rarely needed. For purpose of JEE we will always consider stress to mean Engineering stress. The loading graph above in the left picture shows variation of both types of stress with strain in the rod.

Hooke's Law: In the loading curve of a metal it is seen that for small deformations, stress strain graph is a straight line i.e. stress (σ) $\propto$ strain (ϵ). In this zone where stress is linearly proportional to strain, the material is said to obey Hooke's law.

PROPORTIONAL LIMIT: The point A on stress strain curve after which the material does not obey Hooke's law is called Proportional limit (sometimes also called linear elastic limit).

ELASTIC LIMIT (YIELD POINT): If the material is loaded till point B and then the load is removed, the material completely regains its shape without any permanent elongation. Any extension beyond point B would lead to a permanent elongation upon removal of the load. This point beyond which permanent elongation is inevitable is called Yield Point or Elastic limit.

YIELD STRENGTH: The value of stress in the material at Yield point is called Yield Strength. (σ_y).

PERMANENT SET (PLASTIC DEFORMATION): When the material is loaded beyond its yield strength, upon removal of load, some fractional elongation in the rod remains. This permanent elongation is called Permanent set or plastic deformation (it is typically less than 1%).

ULTIMATE TENSILE STRENGTH: As the specimen is elongated, with increase in strain, the stress keeps increasing up to a point, after this any further increase in strain causes the stress to decrease. The maximum stress that occurs in the loading curve of a material before its fracture is called its ultimate tensile strength. (σ_u). This is represented by point D in the loading curve.

FRACTURE POINT: As we continue to increase the strain in the material, there comes a point where material fractures. This point is called fracture point (and corresponding values of stress and strain are called fracture strain and fracture stress). This is represented by point E in the loading curve.

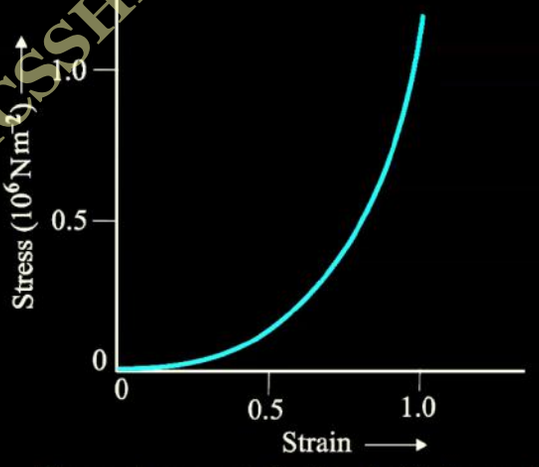
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Ductile Vs Brittle Materials:

If points D and E are far apart in terms of strain difference the material is said to be ductile (Because material can be elongated significantly without fracturing), However if the points D and E are close together in terms of strain values, the material is said to be brittle. For some of the brittle materials the two points may even be coincident i.e fracture stress may be same as the ultimate tensile stress.

Stress Strain Behavior In Various Materials:

Stress-Strain graph may vary greatly from material to material depending on its constituents and structure. Not all materials obey Hooke's law. The adjacent figure shows stress strain curve for Aorta tissue. Here the elastic region is very large and there is no well defined plastic region. The substances which can be stretched to cause large strains without fracture or permanent deformation are called elastomers. We may also plot the stress vs strain graphs for compressive loading and in general the compressive loading curves may not be same as tensile loading curves (for instance, cements have very large compressive strengths but very poor tensile strengths).



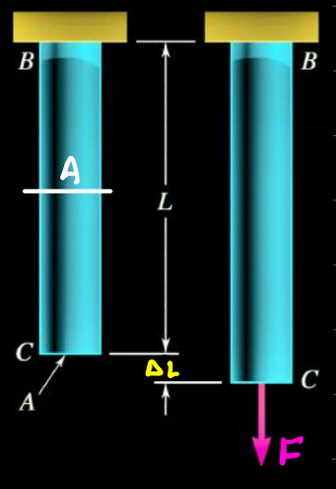
Stress-strain curve for the elastic tissue of Aorta, the large tube (vessel) carrying blood from the heart.

Young's Modulus:

The ratio of stress to strain under axial loading is called Young's Modulus. Mathematically:

$$Y = \sigma/\epsilon$$ (In proportional limit).

Now $\sigma = F/A$ and $\epsilon = \Delta L/L$ thus,



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$$Y = \frac{FL}{A\Delta L}$$

We can also rearrange this equation as:

$$F = \frac{YA}{L}\Delta L$$ thus we see that just like in a spring,

elongation of a rod is directly proportional to the applied force and hence for many purposes, a massless rod of uniform cross section may be treated just like a spring with effective spring constant:

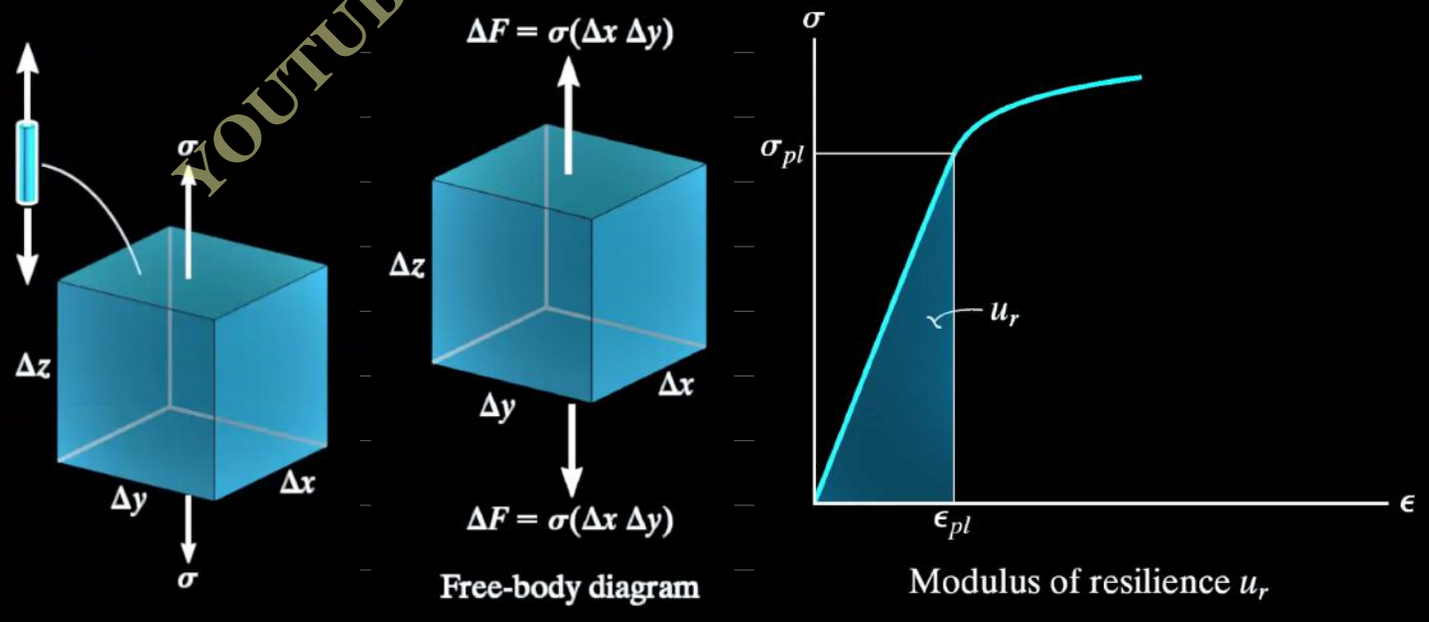
$$K_{eff} = YA/L$$

→ In case the longitudinal stress is not uniform (such as in a massive rod elongating under its own weight) then Young's modulus is equal to the ratio of local stress to local strain. Thus $Y = \sigma/\epsilon = (F/A)/(dy/dx)$ where dy is the infinitesimal elongation in an element of natural length dx.

Thus $\frac{dy}{dx} = \frac{F}{AY}$ and now this equation may be integrated to find total elongation in any portion of the rod. Here F is the tension at the local cross section.

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STRAIN ENERGY:



As a material is deformed by an external load, the load will do external work which in turn will be stored in the material as internal energy. This is known as strain energy. Consider a small element of a material taken from a tension test specimen (figure a). It is subjected to uniaxial stress σ. Thus the force developed by this stress, $\Delta F = \sigma \Delta A = \sigma \Delta x \Delta y$ on the top and bottom faces of the element, which causes the element to undergo a vertical elongation $\epsilon \Delta z$. Here the force is gradually increased from zero to its final value ΔF. Thus work done in the process can be found by multiplying average force with the elongation. Thus energy stored in the element, $\Delta U = \frac{1}{2} \sigma \Delta x \Delta y \times \epsilon \Delta z$ i.e:

$$\Delta U = \frac{1}{2} \sigma \epsilon \Delta V$$ where ΔV is the volume of the element.

For engineering applications often it is more convenient to express strain energy per unit volume (u). Thus :

$$u = \frac{\Delta U}{\Delta V} = \frac{1}{2} \sigma \epsilon$$

Also since $Y = \sigma / \epsilon$, in linear elastic limit we can also write the strain energy density as:

$$u = \frac{1}{2} \sigma \epsilon = \frac{\sigma^2}{2Y} = \frac{1}{2} Y \epsilon^2$$

→ In case of variable stress in any member, the total strain energy can be found by integrating elemental stored energy in various elements. The stress at various sections can be evaluated by considering sectional free body diagrams.

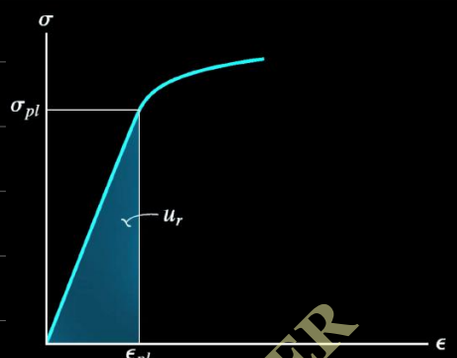
→ Area under the stress strain graph represents the total energy density in the body. This can be readily seen : $dW = F ds$ i.e $dW = \sigma \Delta x \Delta y \Delta z \, d\epsilon = \Delta V \sigma \, d\epsilon$ ($\Delta z \, d\epsilon$ is additional elongation).

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$\Rightarrow \Delta W = \Delta V \int \sigma d\epsilon \Rightarrow u = \int \sigma d\epsilon$ which is nothing but area under σ vs ϵ graph. (Just like with springs stretching by a constant force cause 50% Heat loss).

Modulus Of Resilience:

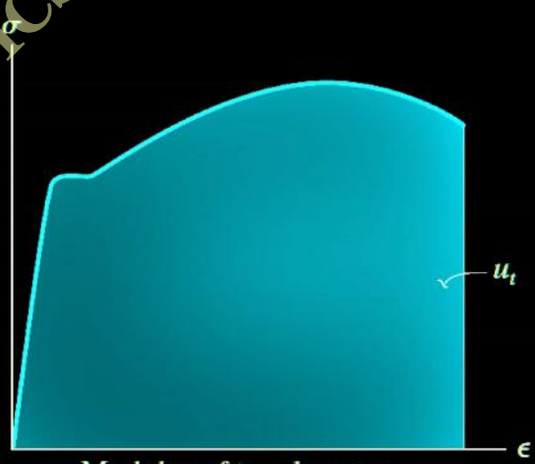
The strain energy density in a material at the point of proportionality limit is called modulus of Resilience. Physically the modulus of resilience largest amount of strain energy per unit volume that a material can take without permanent damage.



Modulus of resilience u_r

Modulus Of Toughness:

Maximum amount of energy per unit volume that a material can absorb before it fractures is called Modulus of Toughness. This is represented by entire area under the stress-strain graph. This can become important considering that overloading of mechanical structures can cause their failure.

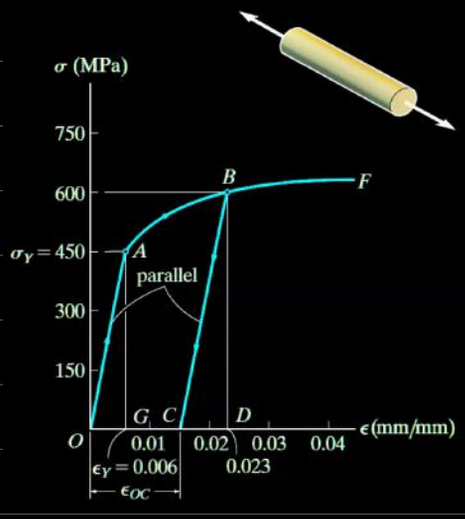


Modulus of toughness u_t

Resilience and toughness can be improved by alloying and it is one of the important study areas of metallurgy and material sciences.

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Example: The stress strain diagram for an aluminium alloy is shown in the figure. If the specimen is stressed to a stress of $\sigma = 600$ MPa, determine the permanent set that remains upon releasing the load. Also comment on what happens to moduli of resilience and toughness? Do they increase, decrease or remain the same.



SOLUTION

Permanent Strain. When the specimen is subjected to the load, it strain hardens until point B is reached on the σ–ϵ diagram. The strain at this point is approximately 0.023 mm/mm. When the load is released, the material behaves by following the straight line BC, which is parallel to line OA. Since both of these lines have the same slope, the strain at point C can be determined analytically. The slope of line OA is the modulus of elasticity, i.e.,

$$E = \frac{450\text{ MPa}}{0.006\text{ mm/mm}} = 75.0\text{ GPa}$$

From triangle CBD, we require

$$E = \frac{BD}{CD}; \qquad 75.0(10^9)\text{ Pa} = \frac{600(10^6)\text{ Pa}}{CD}$$

$$CD = 0.008\text{ mm/mm}$$

This strain represents the amount of *recovered elastic strain*. The permanent set or strain, ϵ_{OC}, is thus

$$\epsilon_{OC} = 0.023\text{ mm/mm} - 0.008\text{ mm/mm}$$

$$= 0.0150\text{ mm/mm} \qquad \textit{Ans.}$$

NOTE: If gage marks on the specimen were originally 50 mm apart, then after the load is *released* these marks will be $50\text{ mm} + (0.0150)(50\text{ mm}) = 50.75\text{ mm}$ apart.

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Modulus of Resilience. Applying definition of resilience, the areas under OAG and CBD in the figure are

$$(u_r)_{\text{initial}} = \frac{1}{2}\sigma_{pl}\epsilon_{pl} = \frac{1}{2}(450\text{ MPa})(0.006\text{ mm/mm})$$

$$= 1.35\text{ MJ/m}^3 \qquad \textit{Ans.}$$

$$(u_r)_{\text{final}} = \frac{1}{2}\sigma_{pl}\epsilon_{pl} = \frac{1}{2}(600\text{ MPa})(0.008\text{ mm/mm})$$

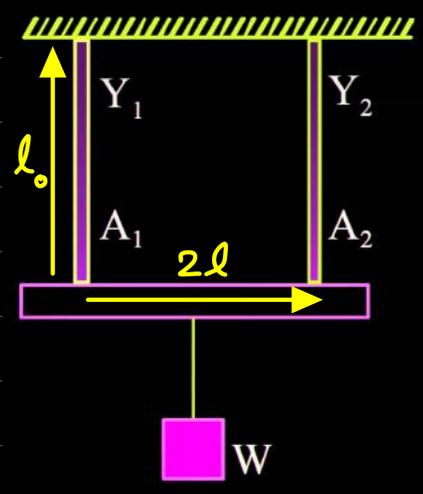
$$= 2.40\text{ MJ/m}^3 \qquad \textit{Ans.}$$

NOTE: By comparison, the effect of strain hardening the material has caused an increase in the modulus of resilience; however, note that the modulus of toughness for the material has decreased, since the area under the original curve, $OABF$, is larger than the area under curve CBF.

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EXAMPLE:

Two thin wires of natural length l_0 each are connected to a massless rod and a weight is suspended from the rod symmetrically. Through what angle does the rod tilt. Assume small elongations.



Solution: Let x_1 and x_2 be the elongations in the strings. Recall that strings behave like springs of constant $k = YA/l$ thus:

$$\left(\frac{Y_1 A_1}{l_0}\right) x_1 + \left(\frac{Y_2 A_2}{l_0}\right) x_2 = W \quad (1)$$

Also by Torque balance, $\left(\frac{Y_1 A_1 x_1}{l_0}\right) \times l = \left(\frac{Y_2 A_2 x_2}{l_0}\right) \times l$ (2)

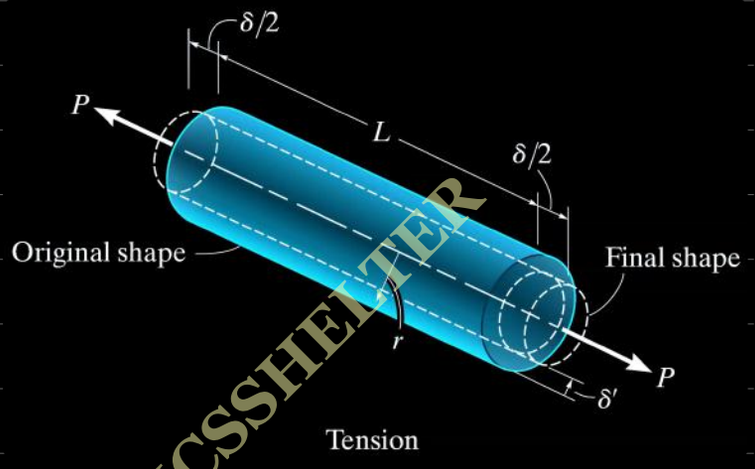
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Solving ① & ② : $x_1 = \frac{l_0 W}{2A_1 Y_1}$, $x_2 = \frac{l_0 W}{2A_2 Y_2}$

Thus angle of tilt : $\theta = (x_2 - x_1)/2l = \frac{W l_0}{4l} \left| \frac{1}{A_2 Y_2} - \frac{1}{A_1 Y_1} \right|$ Ans.

Poisson's Ratio :

When a Bar is subjected to a longitudinal load, not only does is elongate but it also contracts laterally. For example in adjacent figure, a bar of original length L and radius r is subjected to an axial force P. In the process the bar elongates by δ and radius contracts by δ'. Thus the strains in longitudinal and lateral directions are :



$$\epsilon_{long} = \delta/L \quad \text{and} \quad \epsilon_{lat} = -\delta'/r.$$

In the elastic range the ratios of two strains remains constant and the magnitude of ratio of lateral strain to longitudinal strain is called poisson's ratio(ν). Mathematically:

$$\nu = -\frac{\epsilon_{lat}}{\epsilon_{long}}$$

→ It can be theoretically shown that $0 \leq \nu \leq 0.5$. For most non-porous materials, Poisson's ratio lies between 0.25 and 0.35.

→ Do not conserve material volume upon elastic stretching. In general, upon elastic stretching the volume will increase unless $\nu = 0.5$. (With $\nu = 0.5$, volume remains constant).

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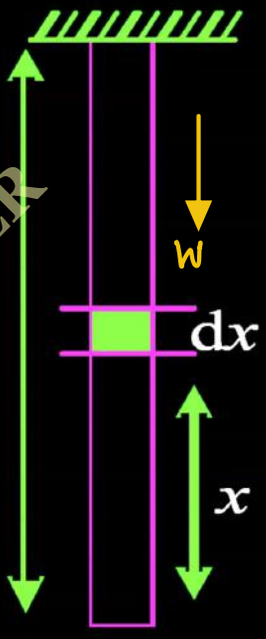
Example: Irodov 1.298:

A copper rod of length l is suspended from the ceiling by one of its ends. Find :

(a) the elongation Δl of the rod due to its own weight;

(b) the relative increment of its volume $\Delta V/V$.

Takes Young's Modulus as Y and Poisson's ratio as σ.



Solution: Consider an element of thickness dx at a distance x from bottom. Here tension T:

$$T(x) = \frac{W}{l}x \Rightarrow \sigma = \frac{W}{Al}x \Rightarrow \epsilon = \frac{\sigma}{Y}$$

$$\Rightarrow \epsilon = \frac{W}{AlY}x = \frac{dy}{dx}$$ where dy is small elongation in the element of natural thickness dx. Thus:

$$\int_0^{\Delta l} dy = \frac{W}{AlY}\int_0^{l} x\,dx \Rightarrow \Delta l = \frac{Wl^2}{2AlY} = \frac{Wl}{2AY}$$

Thus $\Delta l/l = W/2AY$ Ans.

Now for any element volume is given as $V = \pi r^2 l$

$$\Rightarrow \frac{dV}{V} = 2\frac{dr}{r} + \frac{dl}{l} = \frac{dl}{l}[1-2\sigma] \quad \because \sigma = \frac{-dr/r}{dl/l}$$

Since for every element, volumetric strain is simply a scaled version of linear strain, thus for entire rod also $\frac{\Delta l/l}{\Delta V/V} = 1-2\sigma$

$$\Rightarrow \frac{\Delta V}{V} = (1-2\sigma)\frac{\Delta l}{l} = \frac{(1-2\sigma)W}{2AY}$$ Ans.

(Notice that when $\sigma = 0.5$, $\Delta V/V = 0$).

Think: What is the stored strain energy in the rod under its own weight?

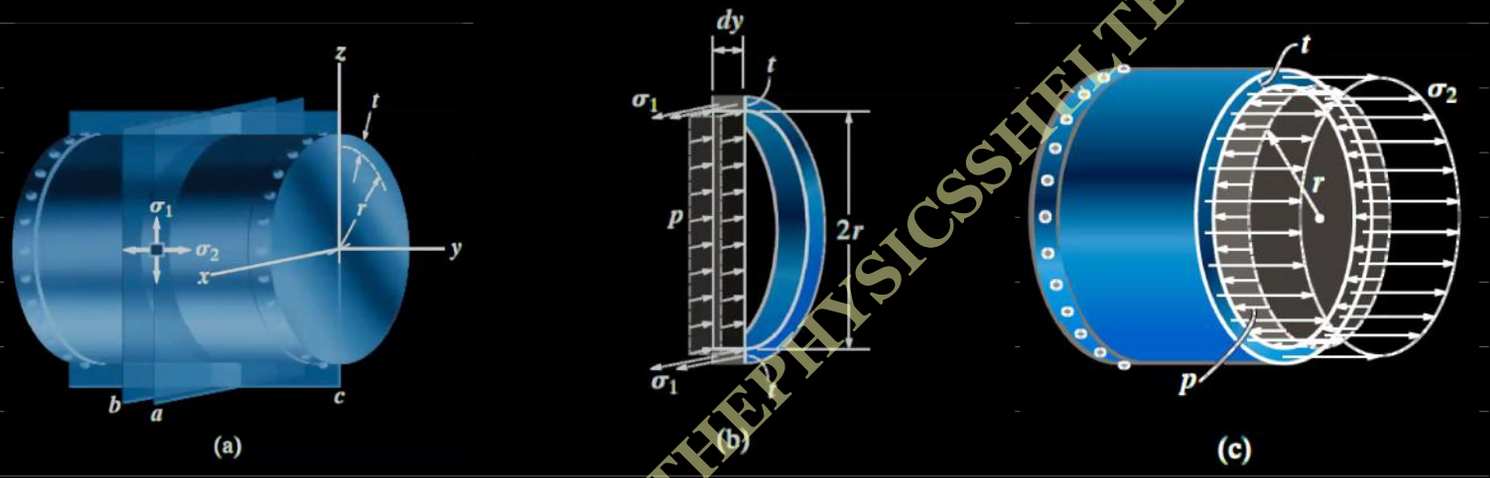
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Solution: $u(x) = \frac{\sigma(x)^2}{2Y} = \left(\frac{Wx}{AL}\right)^2 \cdot \frac{1}{2Y} = \frac{dU}{dV} = \frac{dU}{A\,dx}$

$$\Rightarrow \int_0^U dU = \frac{W^2}{2AYL^2}\int_0^L x^2\,dx \quad \Rightarrow U = \frac{W^2 L}{6AY} \quad \text{Ans.}$$

→ Student is encouraged to verify that this is exactly half of the loss in gravitational PE (the other half is lost as heat).

NORMAL STRESSES IN THIN WALLED PRESSURE VESSELS: (CYLINDRICAL)



When a cylindrical vessel is pressurised, its walls develop stresses. For design analysis we are interested in the normal stresses in its wall along circumferential direction (hoop stress σ_1) and axial direction (longitudinal stress σ_2).

For finding hoop stress we consider a semicircular element of width dy. Let wall thickness be t. Now considering the FBD of this element + air contained between vertical strip and this element (see figure b):

$$\sigma_1 \times t \times dy \times 2 = p \times dy \times 2r$$

$$\Rightarrow \boxed{\sigma_1 = pr/t}$$ ← Hoop stress (circumferential stress).

Now considering FBD of rear half of the vessel and balancing longitudinal forces:

$$p \times \pi r^2 = \sigma_2 \times 2\pi r t$$

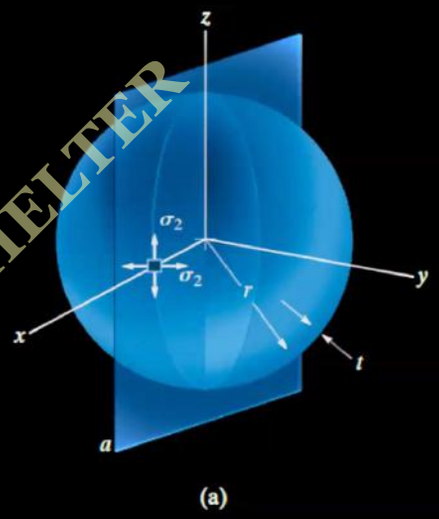
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$\Rightarrow$ $\boxed{\sigma_2 = pr/2t}$ ← Longitudinal stress.

→ We see that hoop stress is double the longitudinal stress thus material failure is more likely to occur due to hoop stress (i.e cracks will appear along length of the vessel).

Normal Stress In Thin Walled Spherical Vessel:

Consider a pressurised spherical vessel having thin walls of thickness t. We are interested in finding the normal stress acting within the walls of the shell. We consider the FBD of left half of the vessel with included air.

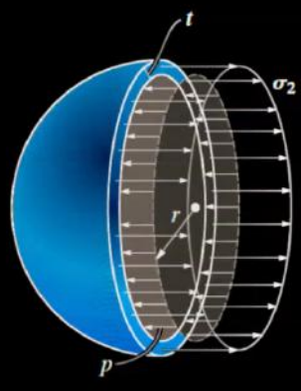


(a)

By horizontal force Balance:

$$p \times \pi r^2 = \sigma_2 \times 2\pi r \times t$$

$\Rightarrow$ $\boxed{\sigma_2 = \frac{pr}{2t}}$ ← Normal stress in spherical shell vessels



(b)

→ We see that this is same as axial stress in cylindrical vessel.

→ From our analysis we can infer that a spherical vessel is capable of handling double the pressure as compared to a cylindrical vessel of same wall thickness.

→ When σ becomes equal to the fracture stress, the vessel will explode. (In case of cylindrical vessel, it is limited by hoop stress).

Thus knowing the fracture stress, we can find the maximum pressure a vessel can withstand.

Fatigue: When pressure vessels are subjected to repeated cycles of pressurising and releasing the pressure, micro-cracks may develop in the material which may extend even at smaller pressures than the theoretical explosion pressures. (That's why on most

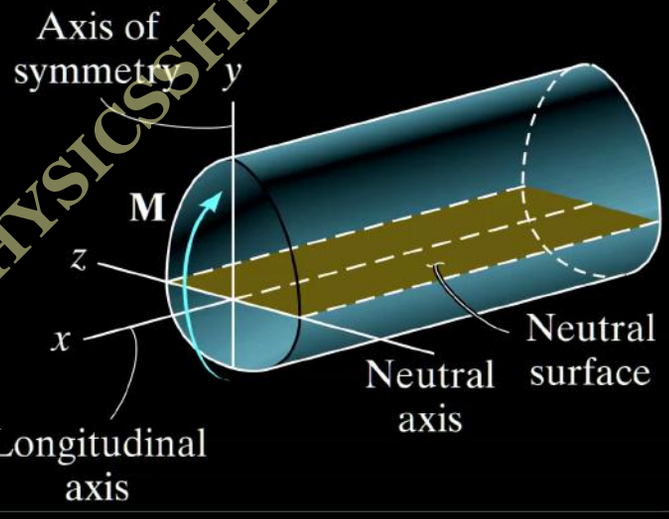
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pressure vessels an expiry date is mentioned beyond which the vessel may not be safe to use).

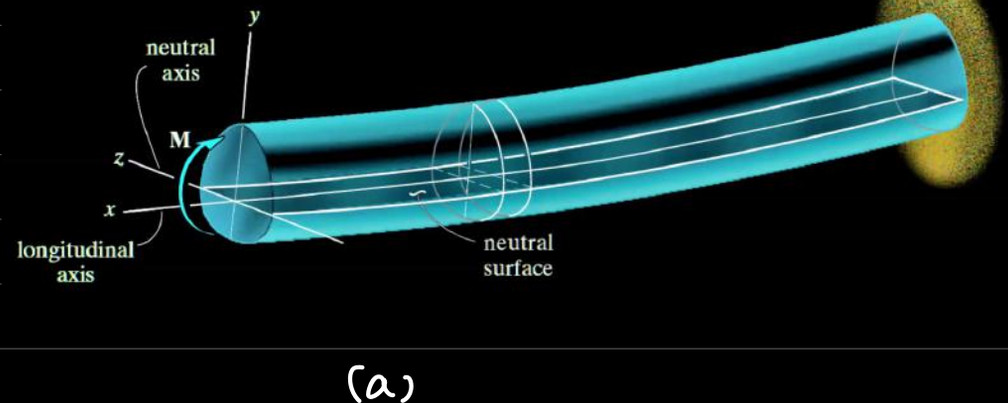
CREEP: When we draw strain vs stress diagram, we see that initially there is one to one relation between stress and strain however in certain materials, (such as lead), the strain not only depends on the load, but also on the duration for which the load is applied. If the load is applied for a long time the strain keeps on becoming larger and larger with time. This behaviour is called creep.

BEAM BENDING: (OPTIONAL)

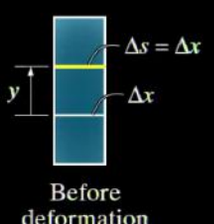
When a beam is subjected to a pure bending moment (external couple), some fibers of the beam develop tensile stresses and some fibers develop compressive stresses. The surface at which fiber are neither in compression nor in tension is called neutral surface and at any cross section, a line on neutral surface along the bending couple is called neutral axis.



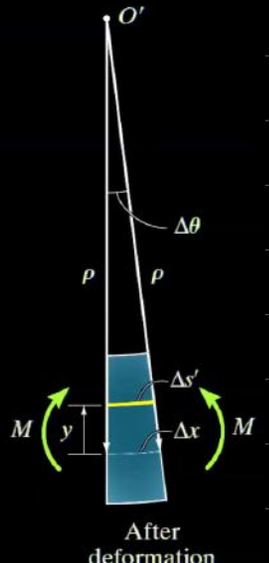
RELATION BETWEEN RADIUS OF CURVATURE AND STRAIN AT GENERAL y:



(a)



(b)



(c)

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Let ρ be the radius of curvature of neutral surface and $\Delta\theta$ be the angle subtended at centre of curvature by a length Δx on the neutral surface (see figure c). Then length of neutral section,

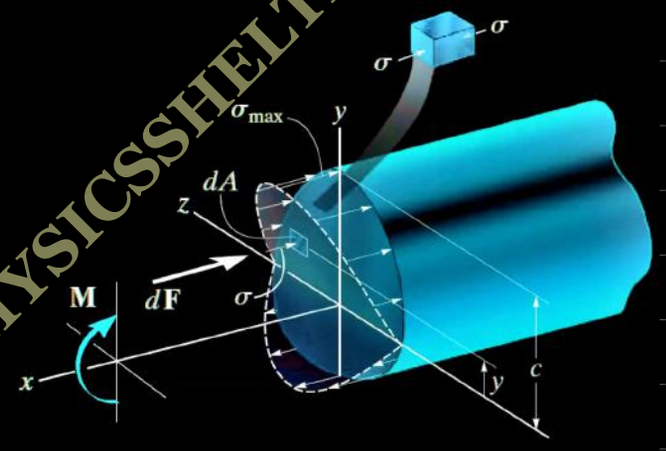
$\Delta x = \rho \Delta\theta$ now for a fiber at a height y above neutral section, the length is $\Delta x' = (\rho - y)\Delta\theta$. Thus its change in length: $\Delta l = -y\Delta\theta$, thus strain, $\epsilon = \frac{\Delta l}{l} = \frac{-y\Delta\theta}{\rho\Delta\theta} = -y/\rho$.

thus $\boxed{\epsilon = -y/\rho}$

Location Of Neutral Axis:

Consider an area dA above neutral axis, the stress here is given as:

$\sigma = Y\epsilon = \frac{-y}{\rho} Y$, thus force associated with this area $dF = \frac{-Y}{\rho} \cdot y\,dA$



Bending stress variation

Now with pure bending couple (no other horizontal forces on beam), Net force on cross section must be zero thus:

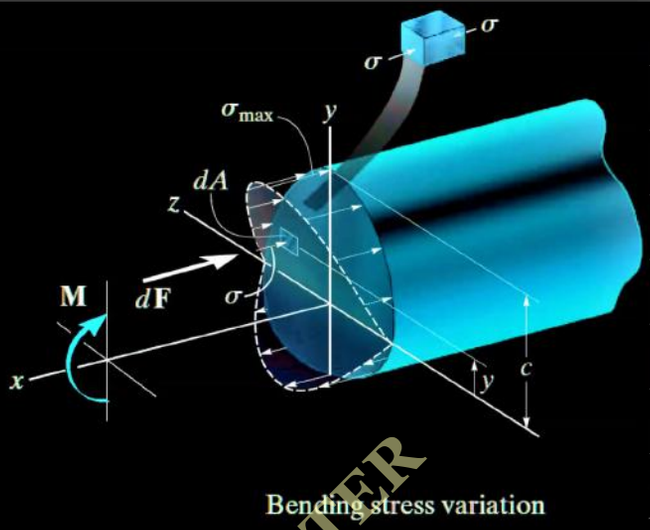
$F = \frac{-Y}{\rho}\int y\,dA = 0$, now recall that a point satisfying $\int y\,dm = 0$ in a lamina is centre of mass of the lamina similarly here $\int y\,dA = 0$ implies that y is being measured from centroid of the area. Thus neutral axis passes through centroid of cross section (so also neutral surface).

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Relation Between Bending Moment And Area Moment Of Inertia:

For equilibrium, external bending moment on any section must be balanced by the couple of the tensile and compressive forces. Now considering couple of stress forces about neutral axis:

$$M = \int \sigma \, dA \, x y$$



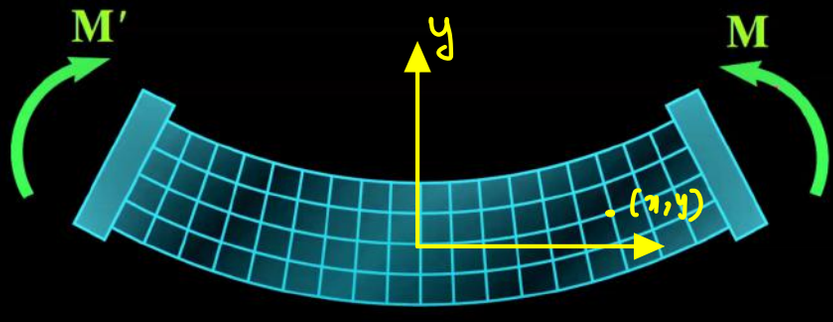
Bending stress variation

$$\Rightarrow M = \int \frac{Y}{\rho} y \, dA \, x y = \frac{Y}{\rho} \int y^2 dA$$

Now $\int y^2 dA$ is nothing but area moment of inertia of cross section about the neutral axis. Thus: $M = \frac{YI}{\rho}$

Euler's Beam Bending Equation (Small Slope Beams):

Recall that radius of curvature of any curve is given as $R = (1+y'^2)^{3/2}/y''$. Now if slope of the curve is small, $y' \approx 0$ and then $R = 1/y''$.



Pure bending with end plates to ensure plane sections remain plane.

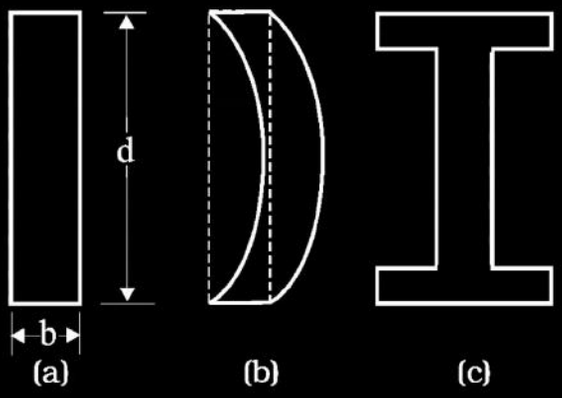
In the previous section we showed that bending moment at a section is given as $M = YI/\rho$ and $\rho = 1/y''$ thus:

$$M = YI\,y'' \Rightarrow \boxed{\frac{d^2y}{dx^2} = \frac{M}{YI}}$$

If we know external bending moment as a function of x, we can use this equation to find shape of the beam.

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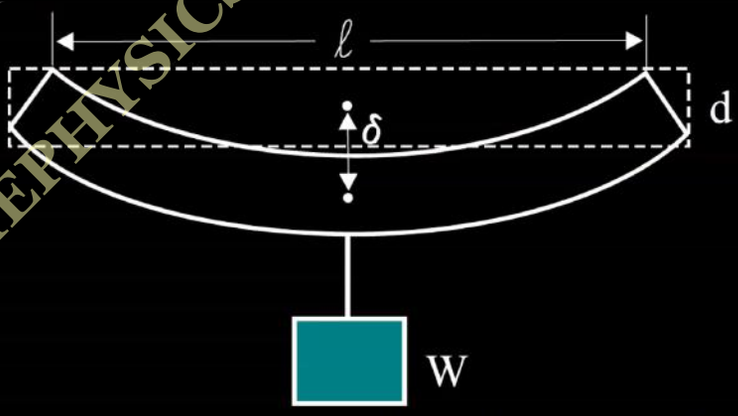
→ The above equation suggests that greater the Area moment of inertia smaller is the bending. That's why often I beams are used as load bearing members because having large mass far away from neutral axis increases the area moment of inertia. (This is also why you sometimes bend a Pizza slice or a roti piece before dipping in to sabzi before eating).



Different cross-sectional shapes of a beam. (a) Rectangular section of a bar; (b) A thin bar and how it can buckle; (c) Commonly used section for a load bearing bar.

Example: Simply Supported Rectangular Section Beam: (NCERT)

Prove that maximum sag in a simply supported rectangular section massless beam under action of a point load W (see figure) is given as $\boxed{\delta = \frac{W l^3}{4 b d^3 Y}}$ Here b is the width of the beam (in to the page), other dimensions are shown in the figure.

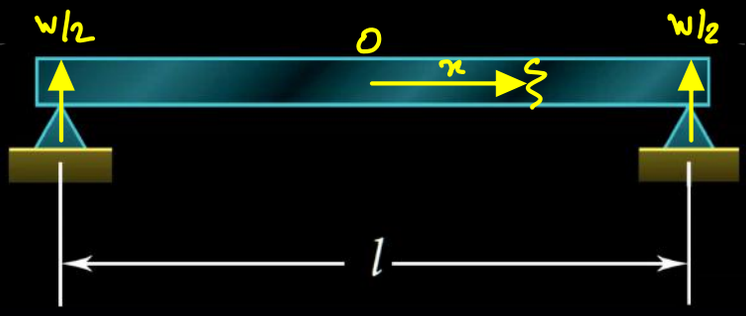


A beam supported at the ends and loaded at the centre.

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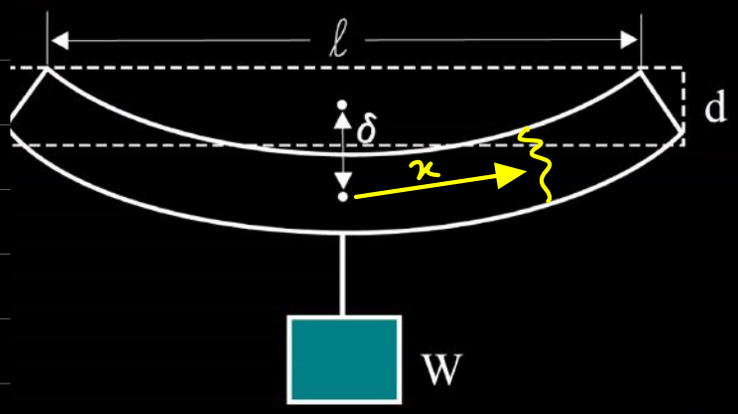
Solution: Consider a section of beam located at a distance x from mid point O as shown. Considering FBD of the beam portion to the right of this section, external torque is:

$$\frac{W}{2} \times \left(\frac{l}{2} - x\right)$$



This torque must be balanced by the torques of compressive and tensile forces about neutral axis. Now using Euler's beam bending equation (small slope)

Also, $\frac{d^2y}{dx^2} = \frac{M}{YI}$



$I = \frac{Ad^2}{12} = \frac{bd^3}{12}$

Thus:

$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{12}{Y\,bd^3} \cdot \frac{W}{2}\left(\frac{l}{2} - x\right)$

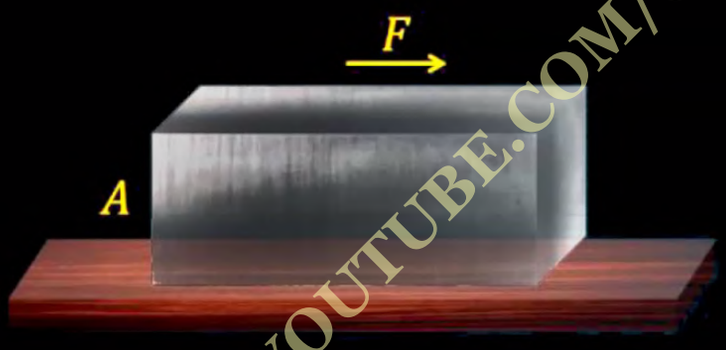
Now @ $x=0$, $\frac{dy}{dx}=0$, integrating wrt x from $x=0$ to $x=x$):

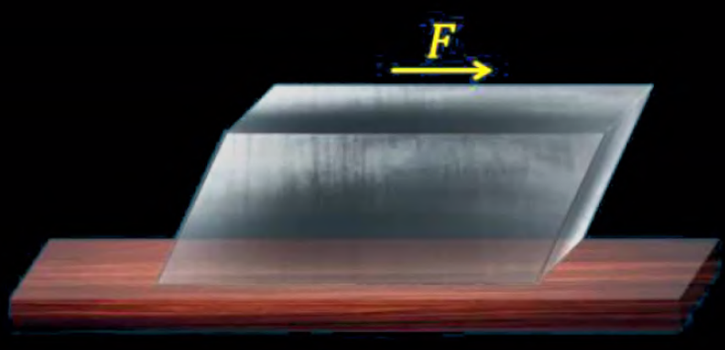
$\frac{dy}{dx} = \frac{3W}{bd^3Y}\left[lx - x^2\right]$ now @ $x=0$, $y=0$, @ $x=\frac{l}{2}$, $y=\delta$

Integrating once again from $x=0$ to $x=l/2$:

$\delta = Wl^3/4bd^3Y$ Ans. QED.

Shear Stress, Shear Strain And Shear Modulus:





Shear Stress: If a force F acts parallel to a surface then the corresponding shear stress is given as $\tau_s = F/A$



Shear Strain: If the shearing force displaces the top surface of a cuboid through a distance x relative to the bottom face then the shear strain is given as $\gamma = \tan\phi = x/l$.

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SHEAR MODULUS: The ratio of shear stress to shear strain is called shear modulus. Thus $\boxed{G = \tau_s/\gamma = (Fl/A\Delta l)}$ $(x \to \Delta l)$

ENERGY DENSITY IN SHEAR DEFORMATION:

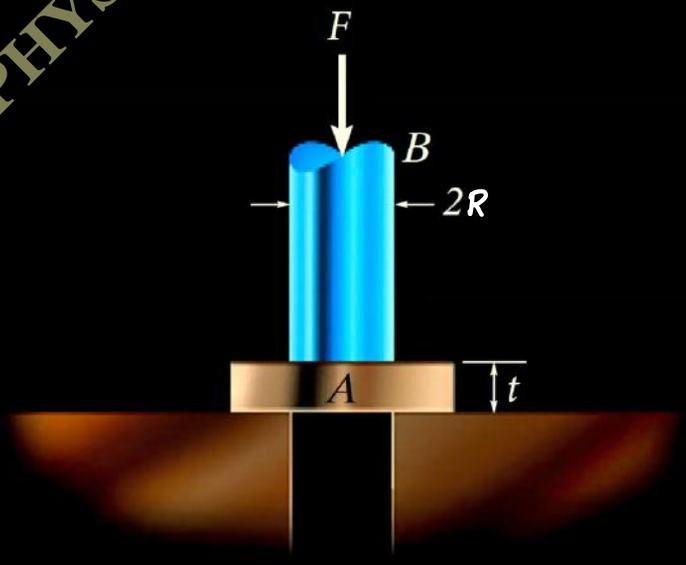
Just like in case of longitudinal loading, under shear loading also, energy density in shear loading is also given as:

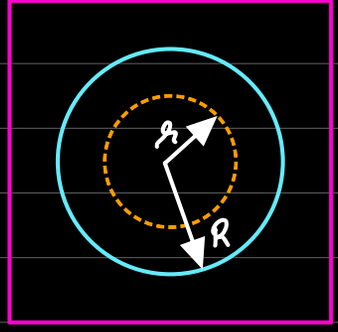
$$\boxed{u = \frac{1}{2} G\gamma^2 = \frac{\tau_s^2}{2G} = \frac{1}{2}\tau_s\gamma}$$

Accordingly, effective spring constant in shear loading is also given as $\boxed{k_{eff} = \frac{GA}{l}}$

SHEAR STRESS VARIATION IN A SHEAR PUNCH:

A shear punch is used for punching holes in sheets. We wish to find shear stress in the sheet as a function of distance r from the punch axis assuming uniform punching pressure. We will consider shearing forces in two regions i.e $r<R$ and $r>R$. (Here by shearing forces, we mean the forces parallel to cylindrical surfaces (imagining the sheet to be made of cylindrical shell layers)). Now let τ_s be the shear stress at a distance r with $r<R$, then $\tau_s \times 2\pi r \times t = \frac{F}{\pi R^2}\cdot \pi r^2$





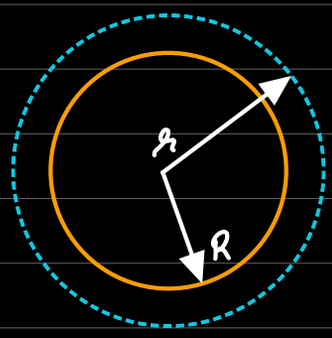
$$\Rightarrow \boxed{\tau_s = \frac{Fr}{2\pi R^2 t}}$$

Thus in the region inside the periphery of the punch, the shear stress increases linearly as one moves outwards towards periphery.

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Now for $r > R$, $F = \tau_s \times 2\pi r t$ thus

$$\tau_s = \frac{F}{2\pi r t}$$

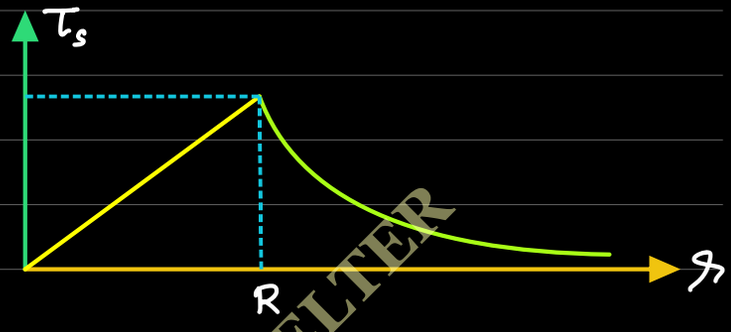


Thus for region outside the periphery of the punch, the shear stress decreases in inverse relation of radius.

The plot of shear stress vs r is shown in adjacent figure.

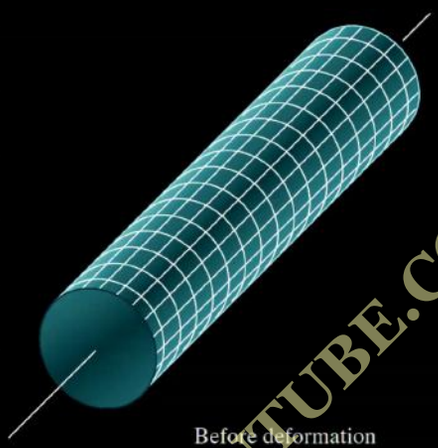
The maximum shear stress occurs at the periphery of the punch.

If τ_0 is the fracture shear stress then force required for punching the hole is given by: $F = 2\pi r t \tau_0$

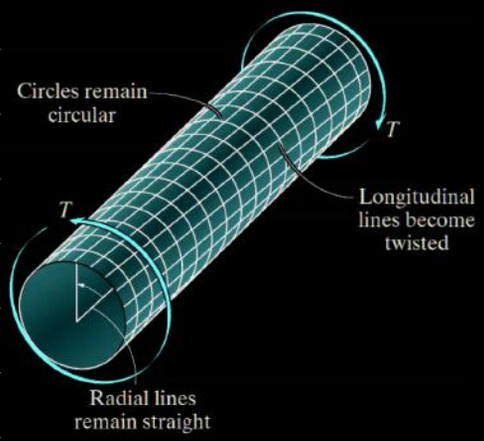


SHEAR STRESS Vs DISTANCE PLOT

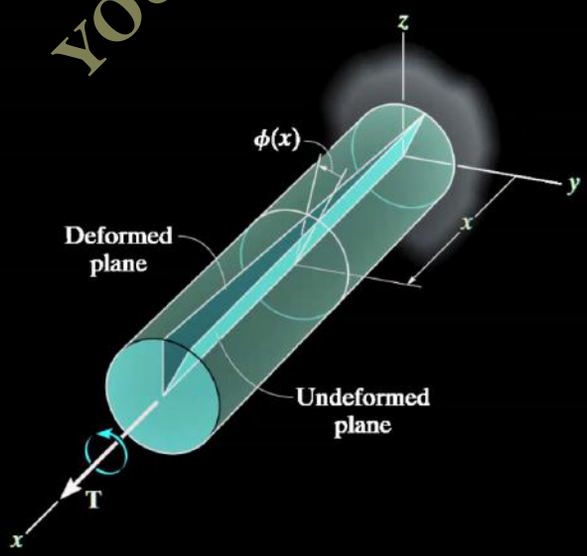
BEAM TORSION:



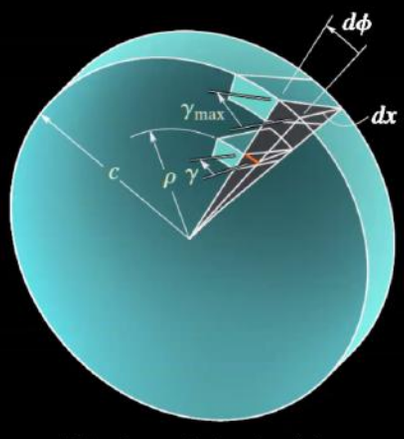
Before deformation
(a)



After deformation
(b)



The angle of twist $\phi(x)$ increases as x increases.
(c)

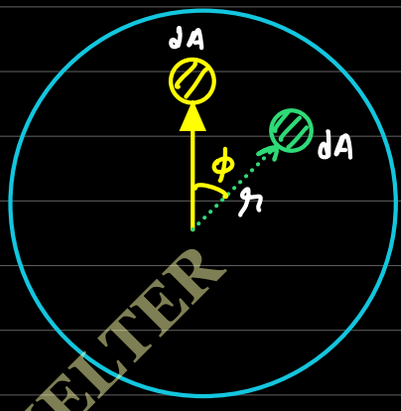


The shear strain at points on the cross section increases linearly with ρ, i.e., $\gamma = (\rho/c)\gamma_{max}$.
(d)

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Imagine a shaft jutting out of a wall. When the front end of the shaft is turned by applying a torque, the longitudinal lines (imagine long fibers) become twisted. The angle of twist is maximum at the front surfaces and zero at the rear face.

Now consider a fiber in the shaft having a cross section area dA. If l is the length of the shaft, then shear strain in the fiber is $\gamma = r\phi/l$ where ϕ is the total twist angle and r is distance of fiber from torsion axis. Accordingly shear stress is then $\tau_s = G\gamma = \frac{Gr\phi}{l}$



Thus shearing force on the fiber is $\tau_s dA = \frac{Gr\phi}{l} dA = dF$

accordingly shearing torque about torsion axis is then

$$dT = r\,dF = \frac{G\phi}{l} r^2 dA$$

Hence total torque is then: $T = \frac{G\phi}{l} \int r^2 dA$

The terms $\int r^2 dA$ is called polar Area moment of inertia and can be readily found using analogy from mass moment of Inertia.

Thus $T = \frac{GI_p}{l}\phi$. Now in a torsional spring the angular deflection is proportional to the applied torque. Thus we see that a shaft behaves just like a torsional spring upon torsion and accordingly effective spring constant can be found as:

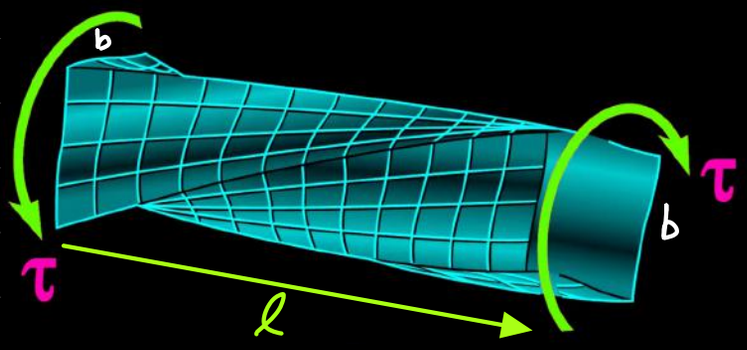
$$T = \frac{GI_p}{l}\phi = C\phi$$

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Thus torsional constant $\boxed{C = \frac{G I_p}{\ell}}$

→ As an example, torsional constant of square section beam shown in adjacent figure is

$$C = \frac{G}{\ell} \cdot \frac{Ab^2}{6} = \frac{G b^4}{6\ell} \quad \text{Ans.}$$



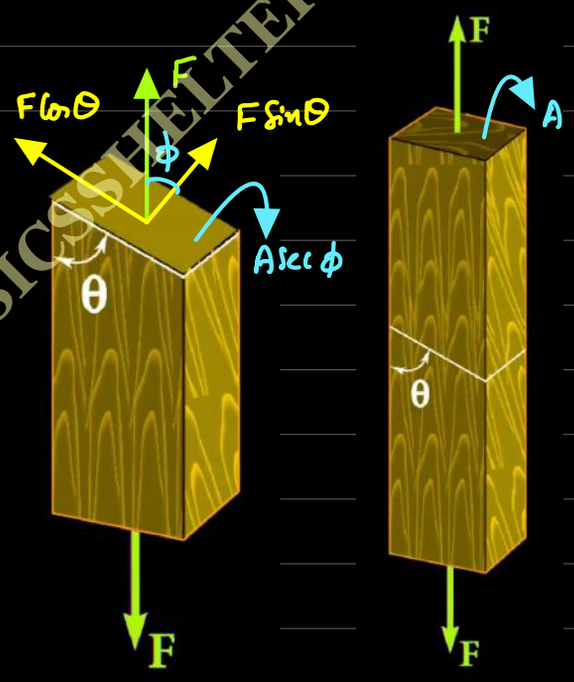
PLANE OF MAXIMUM SHEAR STRESS IN AXIAL LOADING:

Consider a bar of cross sectional area A subjected to a longitudinal load F. If we consider its section at an angle θ as shown, the shear stress on this section is given as:

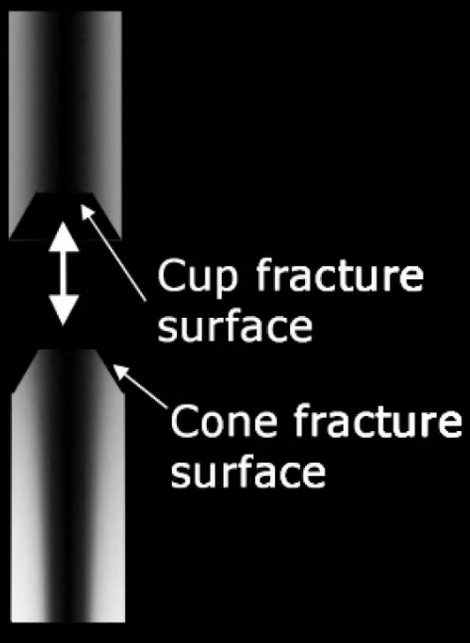
$$\tau_s = \frac{F \cos\theta}{A \sec\phi}, \quad \text{but } \theta + \phi = 90^\circ$$

thus, $\tau_s = \frac{F}{A} \sin\phi \cos\phi = \frac{F}{2A} \sin 2\phi$

Thus maximum shear stress occurs for a plane for which $2\phi = 90^\circ$ i.e $\phi = 45^\circ$.

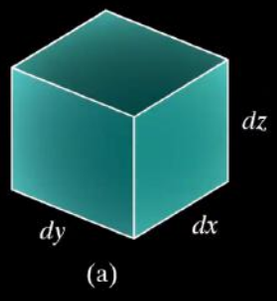


→ In ductile materials typically the material failure is governed by shear strength. Thus in a typical ductile failure, we see a "cup and cone" fracture where failure begins at 45° plane with respect to the loading axis.

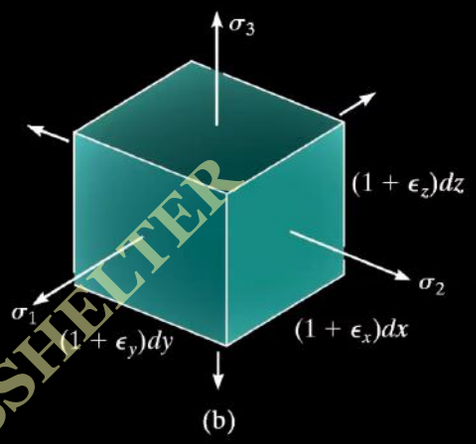


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Multi Axial Loading And Dilatation:



(a)



(b)

Imagine an elastic element being subjected to longitudinal stresses along all the three directions as shown in the figure. If all the stresses are tensile in nature then the volume of the element increases. This is known as dilation. The net volume can be found by superposing the effects of the three loadings considered one at a time. If ϵ_x, ϵ_y and ϵ_z are longitudinal strains along the three directions then:

$$\delta V = (1+\epsilon_x)(1+\epsilon_y)(1+\epsilon_z)\Delta x \Delta y \Delta z - \Delta x \Delta y \Delta z$$

Now expanding and neglecting higher order terms,

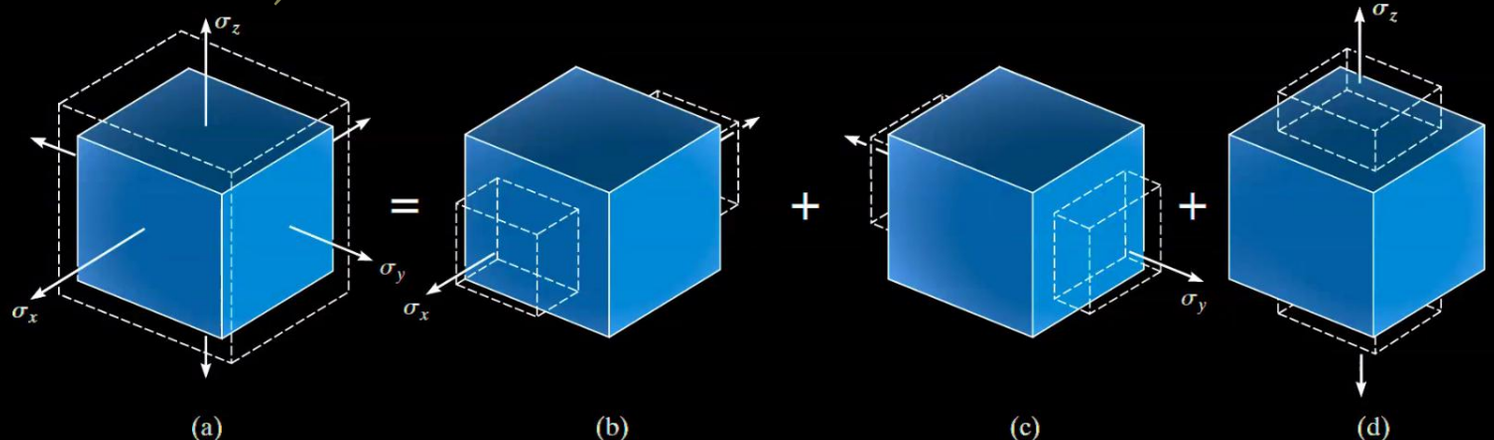
$$\delta V = (\epsilon_x + \epsilon_y + \epsilon_z)\Delta x \Delta y \Delta z = (\epsilon_x + \epsilon_y + \epsilon_z)\Delta V.$$

The change in volume per unit volume is called Volumetric strain or Dilatation (or Bulk strain). Thus, Bulk strain:

$$\epsilon_B = \epsilon_x + \epsilon_y + \epsilon_z = \frac{\delta V}{\Delta V}$$

$\Delta V \rightarrow$ Original volume
$\delta V \rightarrow$ change in volume.

Superposition And Triaxial Stress:



(a) (b) (c) (d)

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If any solid element is subjected to multiple loadings then the net effect of combined loading can be obtained by superposition of the effects of individual loadings (if the loadings are sufficiently small). For instance suppose we wish to find the net strain along x axis due to simultaneous application of σ_x, σ_y and σ_z, we can do this by writing the net strain as:

$\epsilon_x = \epsilon_x' + \epsilon_x'' + \epsilon_x'''$ where ϵ_x' is effect of σ_x and

ϵ_x'' and ϵ_x''' are effects of σ_y and σ_z respectively. Now recall that poisson's ratio is $\nu = -\epsilon_{lat}/\epsilon_{long}$ thus,

$\epsilon_x'' = -\nu\sigma_y/Y$ ($\because$ x direction is lateral to y stress), similarly,

$\epsilon_x''' = -\nu\sigma_z/Y$. Thus:

$$\epsilon_x = \frac{1}{Y}\left[\sigma_x - \nu(\sigma_y + \sigma_z)\right]$$ Similarly,

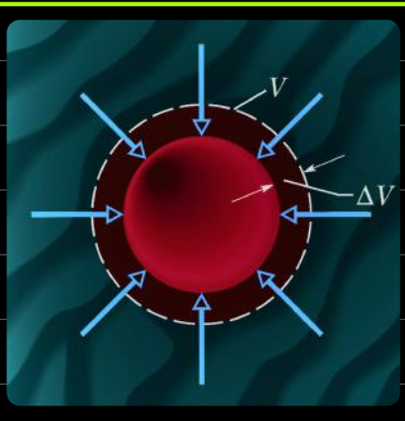
$$\epsilon_y = \frac{1}{Y}\left[\sigma_y - \nu(\sigma_x + \sigma_z)\right]$$ and

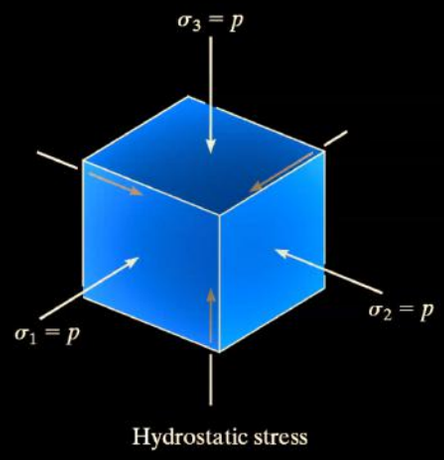
$$\epsilon_z = \frac{1}{Y}\left[\sigma_z - \nu(\sigma_x + \sigma_y)\right].$$

Now recall from the previous section, dilatation $\epsilon_B = \delta V/\Delta V = \epsilon_x + \epsilon_y + \epsilon_z$, Now using expressions for ϵ_x, ϵ_y and ϵ_z derived in this section, dilatation:

$$\epsilon_B = \left(\frac{1-2\nu}{Y}\right)\left[\sigma_x + \sigma_y + \sigma_z\right]$$

Hydrostatic Stress (Bulk Stress):





Hydrostatic stress

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According to Pascal's law when volume element of a material is subjected to a uniform pressure p caused by a static fluid, the pressure will be same in all directions. The shear stresses are not present since the fluid does not flow around the element. Thus here $\sigma_1 = \sigma_2 = \sigma_3 = -p$ ($\because$ compressive stress). This pressure is also known as Bulk stress, or hydrostatic stress (Imagine stress on a concrete block at ocean floor for example).

Bulk Modulus (Volume Modulus Of Elasticity):

The ratio of Hydrostatic stress to corresponding dilatation is called Bulk modulus. Thus:

$$B = \frac{-\Delta P}{(\delta V/V)}$$

where δp is change in pressure and δV is corresponding change in volume. In case of infinitesimal pressure changes, we can also define instantaneous Bulk modulus as

$$B = -dP/(dV/V)$$

Now recall from section on triaxial stress, $\epsilon_B = \left(\frac{1-2\nu}{Y}\right)(\sigma_x + \sigma_y + \sigma_z)$

Thus in case of hydrostatic loading, $\epsilon_B = \left(\frac{1-2\nu}{Y}\right)(-3\Delta P)$

$$\Rightarrow \frac{-\Delta P}{\epsilon_B} = \frac{Y}{3(1-2\nu)} = B$$ thus Bulk modulus is related to young's modulus as:

$$B = \frac{Y}{3(1-2\nu)}$$

Compressibility: The reciprocal of Bulk modulus is called compressibility. Thus compressibility (k):

$$k = \frac{-\Delta V/V}{\Delta P} = \frac{\Delta \rho/\rho}{\Delta P}$$

Where ρ is the density and $\Delta\rho$ is small change in density.

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EXAMPLE: A polytropic process of ideal gas is given as $PV^n = \text{constant}$. Calculate Polytropic Bulk modulus and compressibility.

Solution: $\ln P + n \ln V = \text{Constant} \Rightarrow \frac{dP}{P} + n\frac{dV}{V} = 0$

$\Rightarrow -\frac{dP}{P} = n\left(\frac{dV}{V}\right) \Rightarrow -dP/(dV/V) = nP$ thus polytropic

Bulk modulus of ideal gas is $B_n = nP$. (Now n may be substituted accordingly to desired process. For example $n = 1, \gamma, 0$ respectively for isothermal, adiabatic and Isobaric processes). Compressibility is accordingly $1/nP$ Ans.

RELATION BETWEEN YOUNG'S MODULUS, SHEAR MODULUS & BULK MODULUS:

The relation between Bulk modulus and Young's modulus was derived in a previous section. We can also derive relation between shear modulus and Young's modulus by considering appropriate sectional free body diagrams of a solid however we are stating the result here without the proof (Proof on my other youtube channel). The three moduli are related as:

$$\boxed{Y = 2G(1+\nu) = 3B(1-2\nu)}$$

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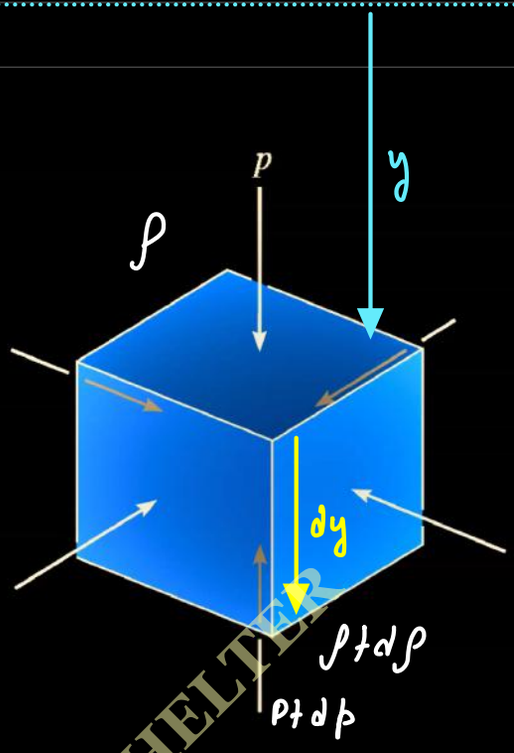
EXAMPLE:

Estimate the change in the density of water in ocean at a depth of 400 m below the surface. The density of water at the surface = 1030 kg m^{-3} and the bulk modulus of water = 2×10^9 N m^{-2}.

Solution: Consider a fluid element at a depth y from ocean surface having a vertical thickness dy, then:

$$dp = \rho g\, dy \quad (1)$$ also bulk modulus

is given as $B = dp/(d\rho/\rho)$ thus:



$$B\frac{d\rho}{\rho} = \rho g\, dy \Rightarrow B\int_{\rho_0}^{\rho}\frac{d\rho}{\rho^2} = g\int_0^y dy$$

$$\Rightarrow B\left[\frac{1}{\rho_0} - \frac{1}{\rho}\right] = gy$$

$$\Rightarrow \rho = \frac{\rho_0 B}{B - \rho_0 g y}$$ Ans now substituting $B = 2\times10^9$, $\rho_0 = 1030\ kg/m^3$, $y = 400$, we get $\rho = 1032.13\ kg/m^3$ Ans.

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